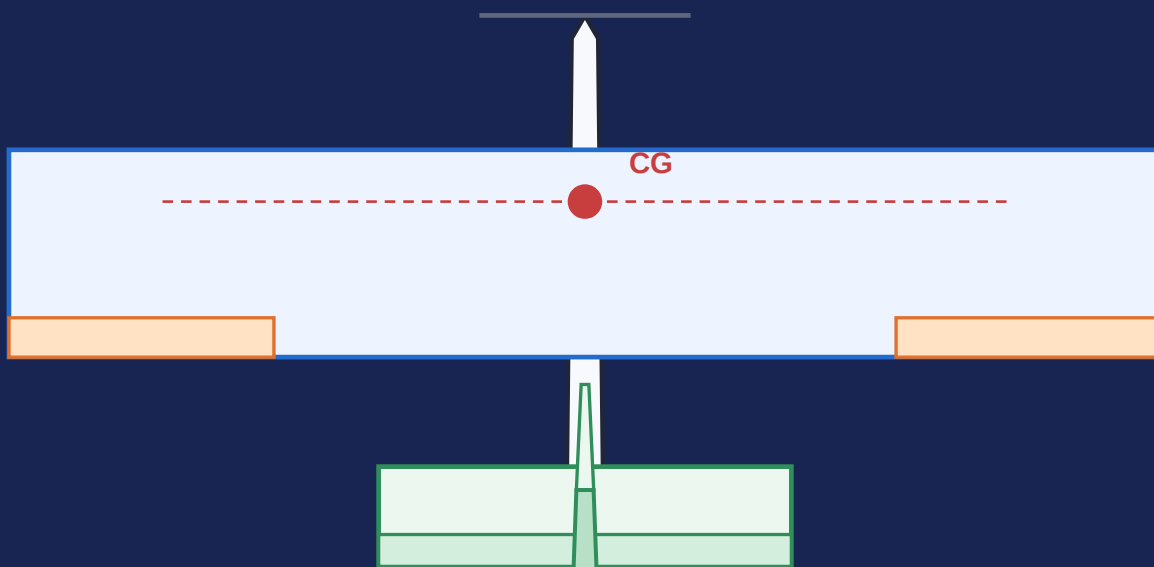


From mission to maiden flight

A practical, calculation-backed guide to sizing, stability, airfoils, structures, propulsion, radio setup, and safe flight testing.



VOLUME 1

Conventional fixed-wing sport and trainer aircraft

Designed for builders who want to understand why the airplane works—not just copy dimensions.

What this manual is—and what it is not

Use it to make sound first-pass decisions, then verify them with testing and experienced help.

THIS MANUAL HELPS YOU

- Turn a mission and target mass into a wing size.
- Size conventional tails using tail-volume coefficients.
- Choose useful starting proportions for control surfaces.
- Build straight, stiff, light structures with reliable linkages.
- Plan a cautious maiden flight and improve the design from evidence.

IT DOES NOT REPLACE

- Local flying-field rules, insurance, or legal requirements.
- Manufacturer limits for batteries, motors, ESCs, servos, and radios.
- An experienced pilot for the maiden of an unproven design.
- Structural analysis for very fast, very large, turbine, or competition aircraft.

THE DESIGN ENVELOPE

The numerical starting points in this volume target ordinary subsonic, conventional-tail, propeller-driven R/C aircraft—especially trainers and sport models. Swept wings, canards, flying wings, biplanes, ducted fans, turbines, and giant scale models need extra analysis.



Safety rule: remove the propeller whenever you are setting up the radio, reversing channels, adjusting endpoints, or working near the power system.

HOW TO READ IT

Green = recommended habit. Orange = design choice. Red = failure or safety risk. Equations are followed by plain-language meaning and worked examples.

The manual map

The front end takes you from requirements through the first flight; the airfoil guide is one chapter in the larger system.

01

Design the system

Mission, forces, controls, wing, tail, CG

02

Understand airfoils

Shape, vocabulary, families, equations

03

Build the structure

Materials, spars, alignment, hinges

04

Choose the systems

Power, propeller, battery, radio, servos

05

Test and improve

Preflight, maiden, trim, diagnosis, repair

! The process is a loop, not a straight line: changing weight changes wing area, power, structure, CG, and often the tail.

The design loop: change one thing, update the rest

A model airplane is a connected system. Every major choice affects several other choices.



! Freeze the mission first. Otherwise the design grows heavier, faster, and more complicated every time a new feature is added.

Write the mission before drawing the airplane

Clear requirements prevent the design from becoming a random collection of appealing features.

FLIGHT TASK

- Trainer, sport, glider, aerobatic, scale, STOL, payload, endurance, or speed?
- Choose one primary mission and at most two secondary goals.

OPERATING SPACE

- Field size, surface, wind, noise limits, transport box, launch method, landing method.
- A small field often demands low wing loading more than high top speed.

TARGETS

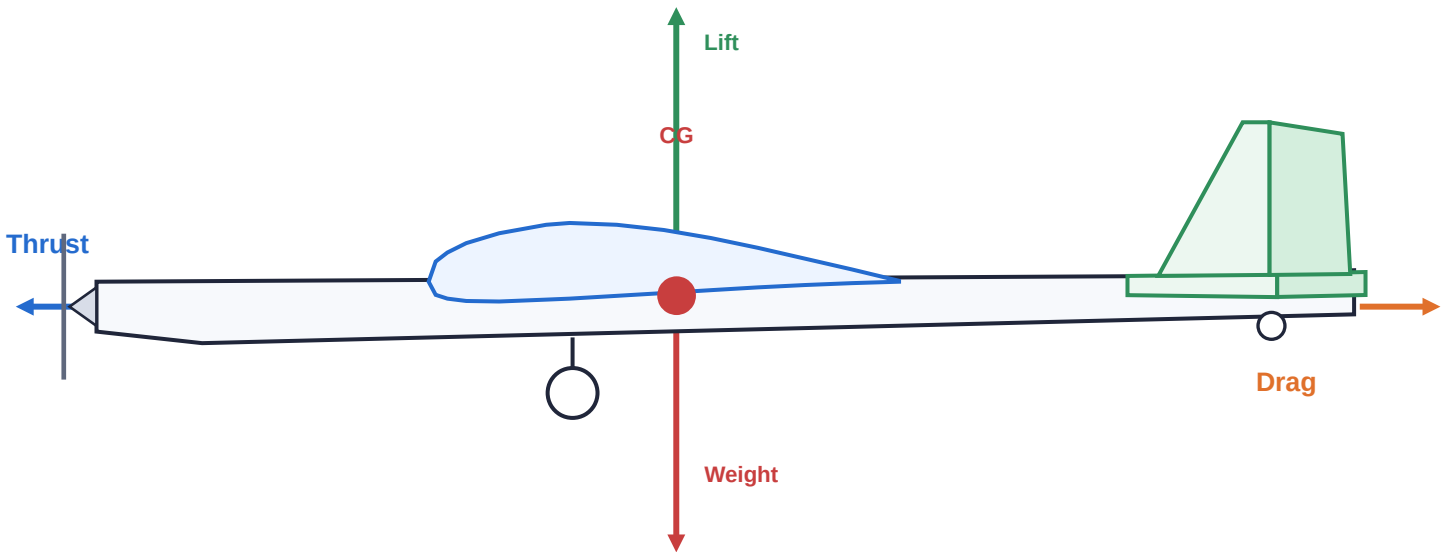
- All-up mass, stall speed, flight time, climb rate, wingspan limit, construction time, and budget.
- Use numbers where possible, not words such as “light” or “fast.”

NON-NEGOTIABLES

- Radio failsafe, battery access, cooling, removable wing, landing gear, camera, tow hook, or float mount.
- Each requirement has mass and drag costs.

Four forces—and the moments they create

An airplane moves because forces act at locations. A force away from the CG also creates a turning moment.



FORCES

- Lift acts roughly perpendicular to the relative wind.
- Weight acts through the CG.
- Thrust and drag act roughly along the flight path.

MOMENTS

- Pitch turns nose up/down.
- Roll rotates wingtip to wingtip.
- Yaw swings the nose left/right.

TRIMMED FLIGHT

Steady flight requires both force balance and moment balance. Lift can equal weight while the airplane still pitches uncontrollably if the moments do not balance about the CG.

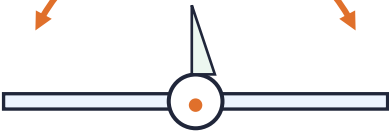
! A tail is not merely a small rear wing. It supplies leverage that balances and damps pitching and yawing motion.

Pitch, roll, and yaw—and the surfaces that control them

The pilot commands rotations. The airplane's stability decides what happens after the command.

ROLL

front view



rotates around the nose-to-tail axis

PITCH

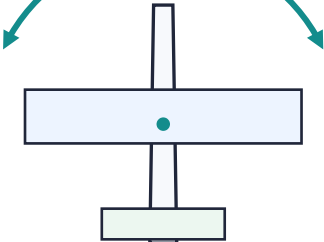
side view



nose tips up / down around the wingtip-to-wingtip axis

YAW

top view



nose swings left / right around the vertical axis

Each view looks straight down the rotation axis (the colored dot) — the motion happens in the plane of the page.

AXIS	MAIN CONTROL	PILOT SEES	STABILIZER
Pitch	Elevator	Nose up or down	Horizontal tail
Roll	Ailerons	Bank left or right	Main wing tips
Yaw	Rudder	Nose left or right	Vertical tail

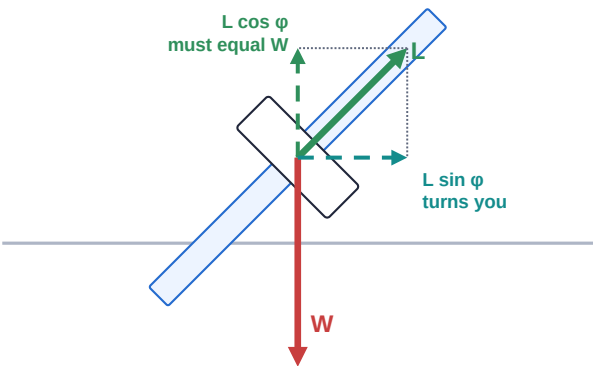


A normal turn is a three-axis event: ailerons bank, elevator maintains the curved flight path, and rudder prevents the airplane from skidding or slipping badly.

Banked turns raise the stall speed

The wing must make more lift in a turn because only part of the lift vector points upward.

45° bank



LOAD FACTOR

$$n = 1 / \cos \varphi$$

φ is bank angle. At 60°, the wing supports about twice the aircraft weight.

TURNING STALL SPEED

$$V_{s,turn} = V_s \sqrt{n}$$

The stall speed increases with the square root of load factor.

BANK	LOAD FACTOR n	STALL-SPEED MULTIPLIER
0°	1.00	1.00 ×
30°	1.15	1.07 ×
45°	1.41	1.19 ×
60°	2.00	1.41 ×



Practical lesson: do not make steep, slow turns near the ground. Add speed before banking, and keep the turn coordinated.

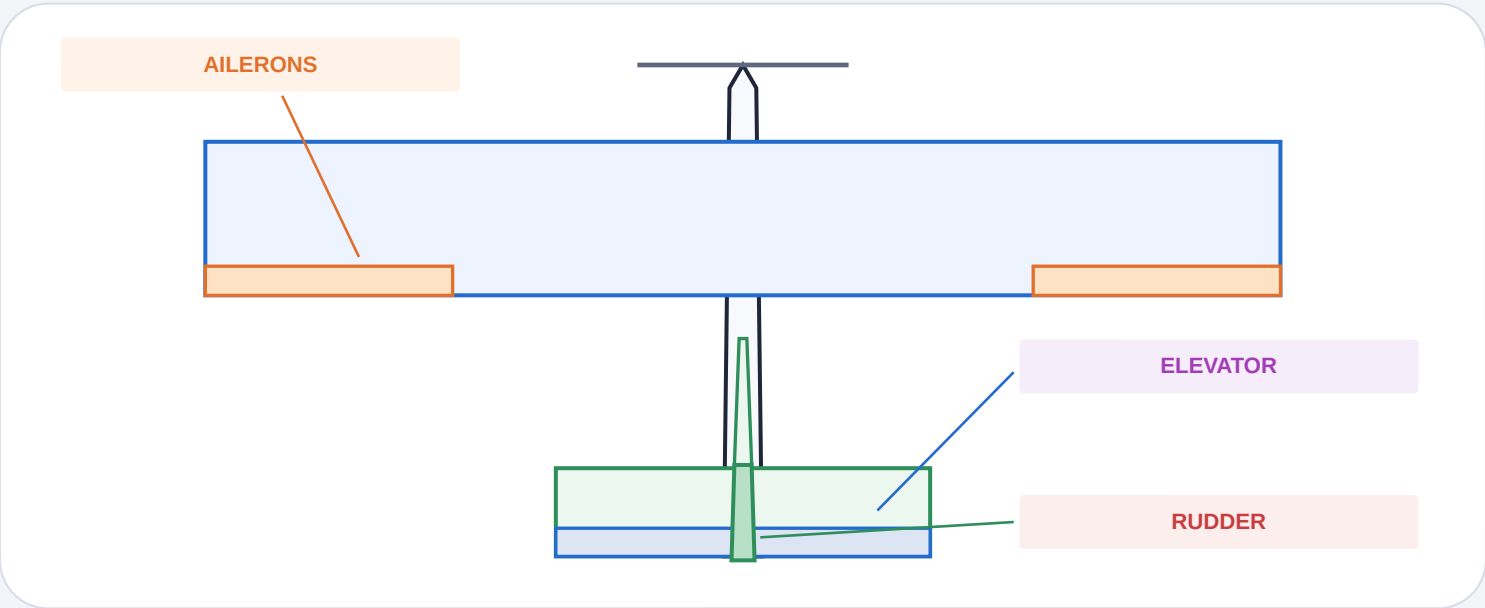


YOUR TURN

At what bank angle does stall speed rise by 25%? Work backwards: $n = 1.25^2 = 1.5625 \rightarrow \cos \varphi = 1/1.5625 = 0.64 \rightarrow \varphi \approx 50^\circ$.

Control surfaces: useful starting proportions

Authority comes from area, lever arm, deflection, airspeed, stiffness, and clean airflow—not area alone.



SURFACE	CHORD	SPAN / HEIGHT	MAIN JOB
Aileron	18–22% of wing chord	Outer 40–50% of each half-wing	Roll
Elevator	25–35% of tail chord	About 80–95% of tail span	Pitch
Rudder	30–40% of fin chord	Most of fin height	Yaw
Flap	20–30% of wing chord	Inner 40–60% of each half-wing	Lift/drag

! These are conventional-trainer starting points, not laws. A fast model needs less deflection; a slow 3D model may use much larger surfaces.

Control throw, rates, expo, and differential

Start modestly. Excess control is harder to manage than slightly too little control.

SETTING	TRAINER START	WHAT IT CHANGES	COMMON MISTAKE
Aileron throw	10–15° each way	Roll rate	Huge travel causes snap-like response
Elevator throw	10–15° each way	Pitch rate / flare	Too much makes CG problems feel worse
Rudder throw	20–25° each way	Yaw / ground steering	Binding at full travel
Flap throw	15–25° takeoff; 30–45° landing	Lift and drag	Deploying too fast without elevator mix
Dual rates	Low for maiden; high available	Maximum travel	Maiden on high rates accidentally
Expo	About 15–30% as a start	Softness near center	Using expo to hide linkage slop
Aileron differential	1.5:1 to 2:1 up/down	Reduces adverse yaw	Equal travel on draggy wings



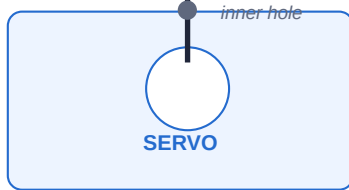
Measure throws at the widest part of the surface. Confirm the radio's expo sign: different brands define positive and negative differently.

Servo and linkage geometry

A rigid, slop-free linkage makes a modest servo work better than a powerful servo connected badly.

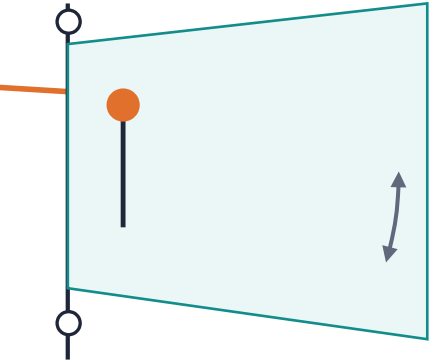
inner servo hole = more force, less travel

outer hole = bigger swing



inner hole

surface swings through this arc



control horn near hinge line

MECHANICAL ADVANTAGE

- Move the pushrod inward on the servo horn for more force and less surface travel.
- Move it outward on the control horn for more force and less travel.
- Keep the control-horn hole close to the hinge line to avoid changing geometry badly.

GOOD INSTALLATION

- Servo arm near 90° to the pushrod at neutral.
- No flexing pushrod, loose clevis, or soft mount.
- No binding at full travel; servo does not buzz continuously.
- Retainers fitted to clevises and wire bends.

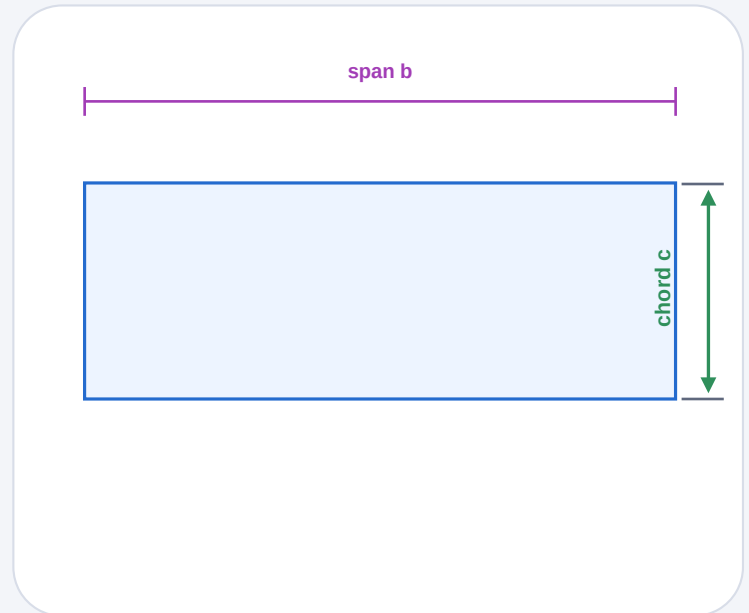


Flutter risk rises quickly with speed. Stiff surfaces, tight hinges, short rigid linkages, and adequate servo torque are structural safety items—not cosmetic details.

How large does the wing need to be?

Start with all-up mass and desired low-speed behavior. Span comes later.

1. Estimate all-up flying mass—including battery, wiring, covering, fasteners, and glue.
2. Choose a target wing loading appropriate to size and mission.
3. Calculate wing area from mass ÷ wing loading.
4. Choose aspect ratio; calculate span and average chord.
5. Check stall speed with an estimated C_{max} .
6. Revisit structure, power, and mass. Repeat until the numbers agree.



AREA FROM WING LOADING

$$S = m \div \text{wing loading}$$

In model units: area in $\text{dm}^2 = \text{mass in grams} \div \text{target g/dm}^2$.

SPAN FROM ASPECT RATIO

$$b = \sqrt{(AR \times S)}$$

After choosing AR, span follows. Average chord is $S \div b$.



Do not select wingspan because it “looks right.” Two aircraft with the same span can have very different areas, loading, stall speeds, and handling.



YOUR TURN

A 2000 g version of your design keeps the same 45 g/dm^2 loading. Find S and the span at AR 6.5. ($S = 44.4 \text{ dm}^2 \rightarrow b = \sqrt{(6.5 \times 0.444)} \approx 1.70 \text{ m}$. Weight grew 33% but span only grew 16% — can you see why in the equations?)

Wing loading: mass carried by each unit of wing area

Lower loading usually means slower flight and gentler landings; higher loading means more speed and energy.

MODEL-BUILDER FORM

$$\text{Wing loading} = \text{mass} / \text{area}$$

Common metric unit: g/dm². Use all-up mass, not bare airframe mass.

SOLVE FOR AREA

$$S = \text{mass} / \text{target loading}$$

This is the fastest first-pass way to size a trainer wing.

SLOW / FLOATY

≈ 25–40

g/dm², size dependent

TRAINER / SPORT

≈ 40–60

g/dm², size dependent

FAST / HEAVY

60+

g/dm²; not a universal boundary

WHY SIZE MATTERS

Simple wing loading does not scale perfectly. Small models operate at lower Reynolds numbers and often need proportionally lighter loading to feel as forgiving as a larger model. Use comparable successful aircraft of similar size as a reality check.



Keep a mass budget. Reserve roughly 10–15% for items you forgot. Weight added late forces faster flight, stronger structure, and often a larger power system.

Stall speed: the speed where the wing runs out of usable lift

The equation links weight, area, air density, and maximum lift coefficient.

STALL-SPEED ESTIMATE

$$V_s = \sqrt{2 W / (\rho S C_{lmax})}$$

W is weight in newtons, ρ is air density, S is wing area, and C_{lmax} is the wing's maximum lift coefficient.

WEIGHT W

More weight raises stall speed. Doubling weight raises V_s by $\sqrt{2}$, not by 2.

AREA S

More wing area lowers stall speed. It also adds structure and drag.

C_{lMAX}

Airfoil, roughness, Reynolds number, flap setting, planform, and stall pattern all affect it.

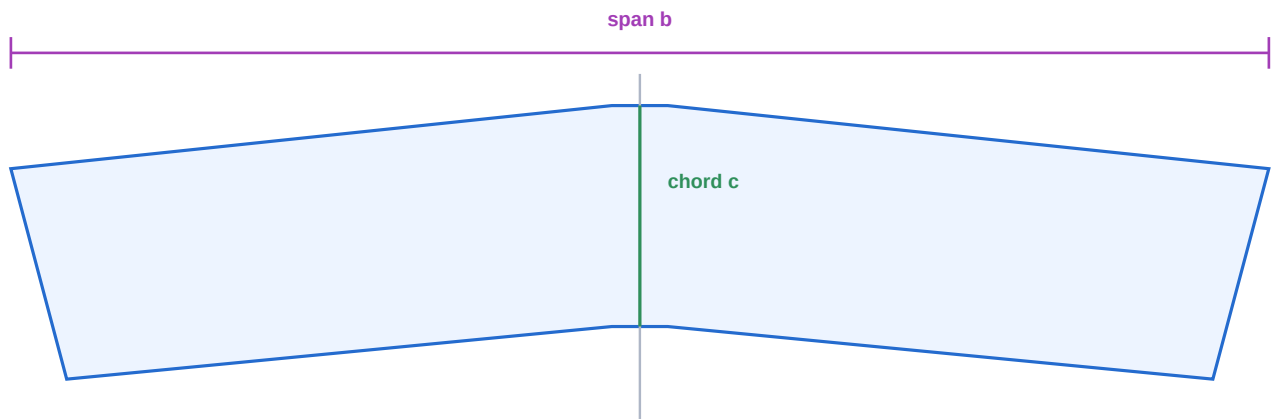
SPEED MARGIN

A practical approach is often around $1.3 V_s$ in calm air, then adjusted for gusts and pilot skill.

 C_{lmax} is usually the least certain input. Treat the result as an estimate and validate with conservative flight testing at safe altitude.

Span, chord, and aspect ratio

Wing area sets the basic lift requirement; aspect ratio decides how that area is distributed.



ASPECT RATIO

$$AR = b^2 / S$$

Long, narrow wings have high AR. Short, broad wings have low AR.

RECTANGULAR WING

$$S = b \times c$$

For a rectangle, average chord is simply area divided by span.

HIGHER AR

- Lower induced drag for a given lift.
- Efficient gliding.
- Longer spar and greater bending challenge.

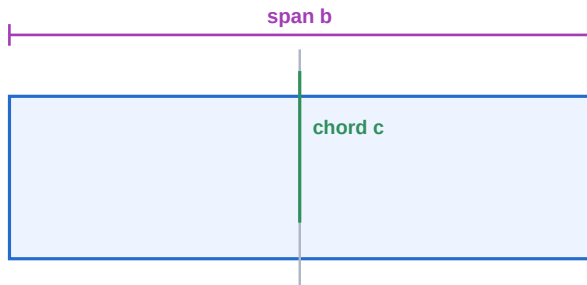
LOWER AR

- Compact and often agile.
- More induced drag.
- Usually easier to make stiff.

Planform, taper, and sweep

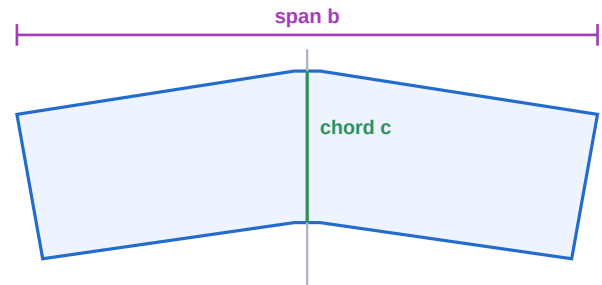
The outline of the wing affects stall progression, structure, efficiency, and control placement.

RECTANGULAR



Easy to build; friendly stall; excellent first design.

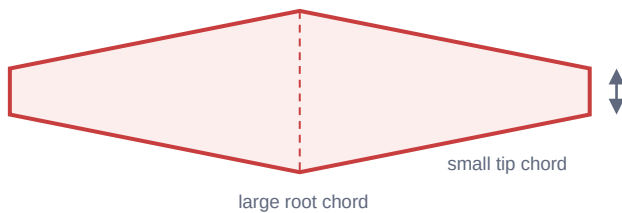
MILD TAPER



Less tip area and drag; keep taper modest for a trainer.

STRONG TAPER

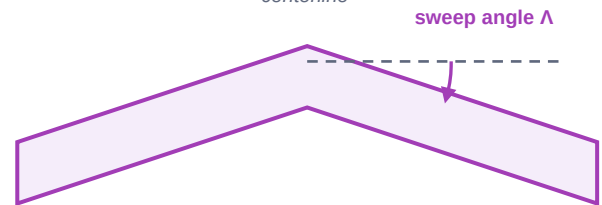
chord shrinks toward the tip — leading and trailing edges both angle in, no sweep



Can tip-stall abruptly without washout or airfoil changes.

SWEPT

chord stays the same — the whole panel angles back from the centerline



Moves aerodynamic center and complicates CG/MAC reasoning.

Dihedral, washout, incidence, and decalage

These angles change how the airplane returns to level, stalls, and trims.

DIHEDRAL — tips rise (front view)



Tips sit HIGHER than the root. In a sideslip, the low wing meets the air at a higher angle, lifts more, and rolls the plane level.

INCIDENCE — how the wing is mounted



A construction angle set in the jig: the wing chord sits a few degrees nose-up relative to the fuselage datum. It never changes in flight — angle of attack does.

WASHOUT — tip twisted nose-down



The root reaches stall angle first while the tip — and its aileron — is still flying. A stall becomes a gentle nod, not a snap roll.

DECALAGE — wing minus tail incidence



The difference between the two incidences. It sets how hard the tail must push to hold the wing at its working angle — i.e., the elevator trim.

INCIDENCE

The fixed angle of a wing or tail relative to a fuselage reference line. It is a construction setting, not the same as angle of attack in flight.

DECALAGE

The angular difference between the wing and horizontal tail reference lines. It influences the elevator position required for trim.

TRAINER START

- Rudder/elevator-only: noticeable dihedral.
- Aileron trainer: moderate dihedral.
- Washout: often about 1–2° at each tip.

BUILD RISK

- Both panels must match.
- Accidental twist is not washout.
- Covering can pull a straight wing out of shape.

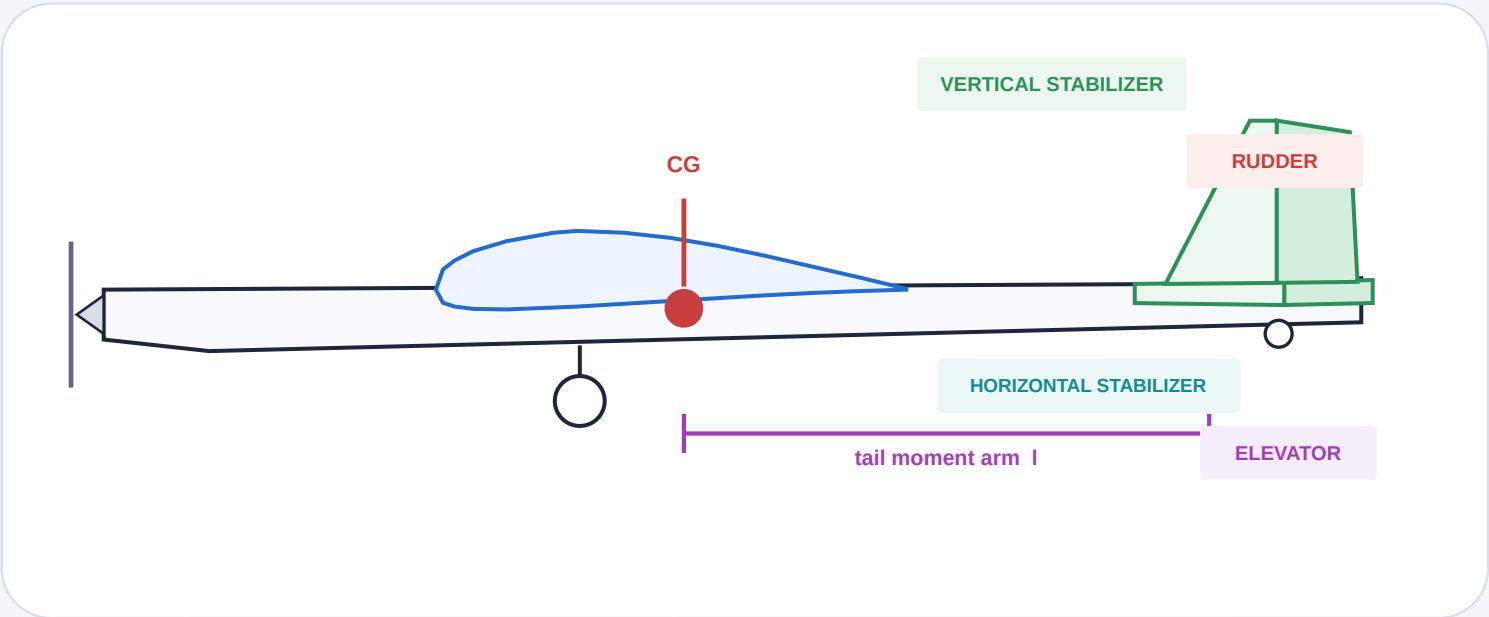
YOUR TURN



You measure the left wingtip at +1.4° washout and the right tip at +0.6°. Is that washout, warp, or both? Which wing drops first near the stall — and what does the plane do next?

Tail anatomy: stabilizers and control surfaces

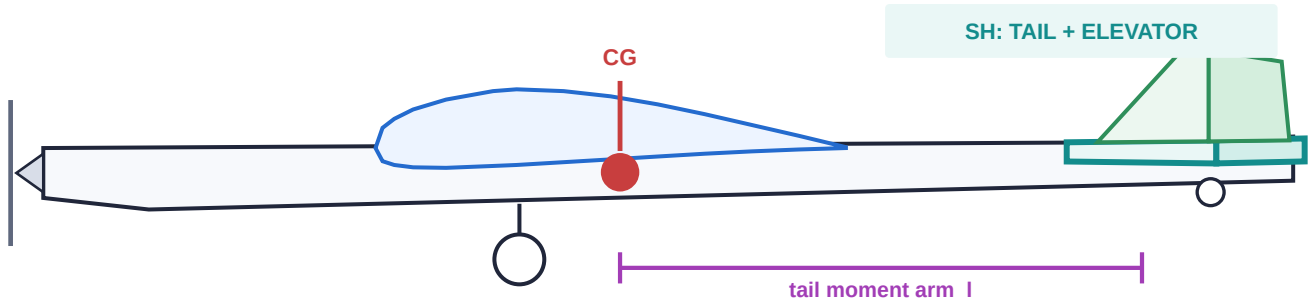
The complete tail assembly is the empennage. Fixed surfaces stabilize; moving surfaces command.



PART	WHY IT EXISTS	CONTROL / CONSEQUENCE
Horizontal stabilizer	Pitch stability and damping	Elevator
Vertical stabilizer / fin	Yaw stability and damping	Rudder
Tail moment arm	Leverage behind the CG	Allows smaller tail for same volume

Horizontal-tail volume: area × leverage for pitch stability

A small tail far behind the CG can be as effective as a larger tail mounted close.



Pitch leverage = horizontal-tail area × distance behind CG

HORIZONTAL TAIL VOLUME

$$V_h = S_h l_h / (S \bar{c})$$

S_h = horizontal-tail area, l_h = CG-to-tail arm, S = wing area, $\bar{c}$ = mean aerodynamic chord.

SOLVE FOR TAIL AREA

$$S_h = V_h S \bar{c} / l_h$$

Choose V_h , then calculate area. A longer fuselage reduces required tail area.

BROAD RANGE

0.30–0.60

conventional aircraft

TRAINER START

0.50–0.60

forgiving first pass

EXAMPLE

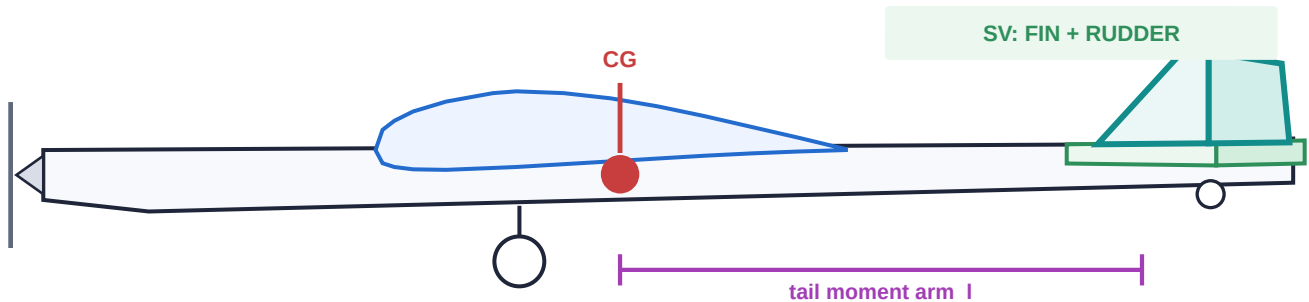
640 cm²

1.5 kg trainer below

! Compare the result with successful aircraft of similar configuration. Volume coefficients are screening tools, not proof of stability.

Vertical-tail volume: area × leverage for yaw stability

The fin resists unwanted yaw and damps side-to-side motion; the rudder commands yaw.



Yaw leverage = vertical-tail area × distance behind CG

VERTICAL TAIL VOLUME

$$V_v = S_v l_v / (S b)$$

S_v = fin-plus-rudder area, l_v = CG-to-fin arm, S = wing area, b = wing span.

SOLVE FOR FIN AREA

$$S_v = V_v S b / l_v$$

More side area ahead of the CG—large canopy, floats, nose—may require more fin behind the CG.

BROAD RANGE

0.02–0.05

conventional aircraft

TRAINER START

0.035–0.05

good yaw damping

EXAMPLE

300 cm²

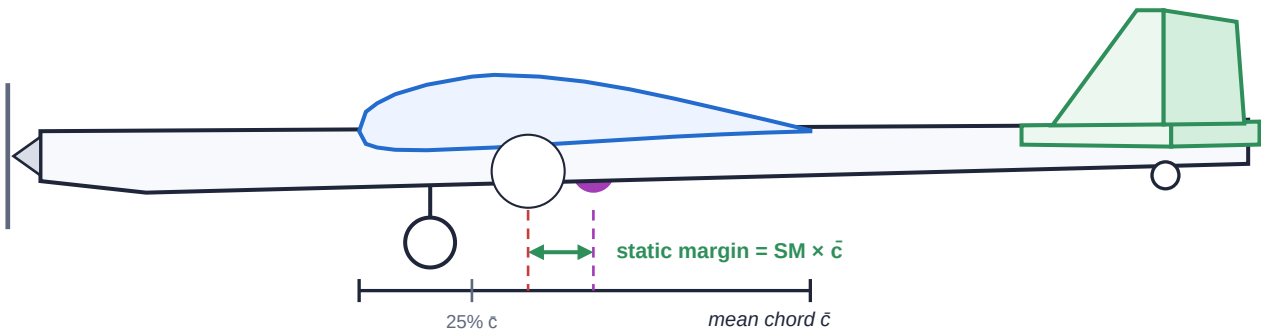
1.5 kg trainer below



An undersized fin can produce weak directional stability and Dutch-roll-like wandering. A very large fin can make crosswind handling and structure harder.

Center of gravity, neutral point, and static margin

CG position can turn a well-built airplane from calm to uncontrollable.



CG ahead of NP = stable. Spacing exaggerated for clarity; 25% $\bar{c}$ is a common quick-start CG for unswept trainers.

STATIC MARGIN

$$\text{SM} = (x_{\text{NP}} - x_{\text{CG}}) / \bar{c}$$

Positive static margin means the CG is ahead of the neutral point.

MAIDEN TARGET

$\approx 10\text{--}15\%$

conservative conventional design

CG TOO FAR FORWARD

- Very stable but nose-heavy.
- Needs more elevator; weak flare.
- Higher trim drag and landing speed.

CG TOO FAR AFT

- Pitch-sensitive and hard to trim.
- May oscillate or pitch up.
- Can become unrecoverably unstable.

YOUR TURN

?

$\bar{c} = 226$ mm and you balance the model 12 mm ahead of the neutral point. What is the static margin, and is it enough for a maiden? ($12 / 226 \approx 5\%$ — too little; move the CG forward.)

Stability and control authority are different

A stable aircraft returns toward trimmed flight; an authoritative control can move it strongly. You need both in the right amounts.

STATIC STABILITY

- Initial tendency after a disturbance.
- Positive pitch stability means a nose-up disturbance creates a restoring tendency.

DAMPING

- How quickly oscillations die out.
- Tails, fuselage, wing, and speed all contribute.

CONTROL AUTHORITY

- Ability to create roll, pitch, or yaw moment.
- Depends on surface size, lever arm, deflection, dynamic pressure, and stiffness.

TRIM

- A steady control position where moments balance.
- A plane can be stable yet badly trimmed—or controllable yet unstable.

Step 1: build a realistic mass budget

The example is a 1.5 kg electric trainer designed for calm-to-moderate field flying.

ITEM	BUDGET	NOTE
Airframe structure	480 g	Wing, fuselage, tail, landing gear
Motor + prop + mount	180 g	Approx. 220 W class
ESC + wiring	55 g	With cooling margin
Battery	185 g	3S 2200 mAh example
Receiver + 4 servos	115 g	Includes leads and hardware
Covering / finish	85 g	Often underestimated
Fasteners / glue / misc.	100 g	Hinges, horns, wheels, connectors
Design reserve	300 g	20% early-design reserve
TARGET ALL-UP MASS	1500 g	The number used for sizing



At each design review, replace estimates with measured masses. If the total grows, recalculate wing loading, stall speed, power, and CG.

Step 2: size the wing

Target: forgiving 1.5 kg trainer with approximately 45 g/dm² wing loading and AR 6.5.

MASS

1500 g

all-up

TARGET LOADING

45 g/dm²

trainer target

ASPECT RATIO

6.5

moderate

AREA

$$S = 1500 / 45 = 33.3 \text{ dm}^2$$

$$33.3 \text{ dm}^2 = 0.333 \text{ m}^2.$$

SPAN

$$b = \sqrt{(6.5 \times 0.333)} = 1.47 \text{ m}$$

Choose about 1470 mm.

AVG. CHORD

$$c = 0.333 / 1.47 = 0.226 \text{ m}$$

A rectangular wing uses about 226 mm chord.

BUILD DECISION

- Use a rectangular or mildly tapered planform for predictable construction and stall behavior.
- Estimated $C_{\text{max}} = 1.1$ gives $V_s \approx 8.1 \text{ m/s}$ (29 km/h); a calm-air approach near $1.3 V_s$ is about 10.5 m/s.
- Keep the surface smooth and the leading edge accurate; low-Reynolds-number performance is sensitive to roughness and shape errors.

YOUR TURN

Rerun this whole page for a 2 kg sport model: 60 g/dm², AR 7, C_{max} 1.0. Find S, b, c, and V_s . (Answers: $S = 33.3 \text{ dm}^2$, $b \approx 1.53 \text{ m}$, $c \approx 218 \text{ mm}$, $V_s \approx 9.8 \text{ m/s}$ — same wing area as the trainer, yet it stalls 1.7 m/s faster. Why?)

?

Step 3: size the tail and controls

Use a 650 mm tail arm, $V_h = 0.55$, $V_v = 0.04$, and the wing dimensions from the previous page.

WING AREA 0.333 m² S	WING SPAN 1.47 m b	MEAN CHORD 0.226 m $\bar{c}$	TAIL ARM 0.65 m l
--	---	--	--

H-TAIL

$S_h \approx 0.064 \text{ m}^2$

About 640 cm²; for example, 470 × 135 mm average.

V-TAIL

$S_v \approx 0.030 \text{ m}^2$

About 300 cm²; for example, 240 mm high × 125 mm average chord.

BUILD DECISION

- Elevator: about 30% of horizontal-tail chord—roughly 40 mm—and most of the tail span.
- Rudder: about 35–40% of fin chord and most of the fin height.
- Ailerons: about 45 mm chord, roughly 330–370 mm span each, with 1.5:1 to 2:1 up/down differential.
- Begin with conservative CG and modest throws; move aft or increase travel only after evidence from safe flights.

Step 4: power system and maiden configuration

The objective is a confident climb—not unlimited vertical performance.

POWER TARGET

130–160 W/kg

trainer range

REQUIRED INPUT

195–240 W

for 1.5 kg

BATTERY EXAMPLE

3S 2200 mAh

use correct C rating

POWER

$1.5\text{ kg} \times 150\text{ W/kg} \approx 225\text{ W}$

Choose motor, propeller, and ESC as a tested combination.

ROUGH ENDURANCE

$0.8 \times 2.2\text{ Ah} / 12\text{ A} \times 60 \approx 8.8\text{ min}$

Actual current changes continuously; use a timer and telemetry if available.

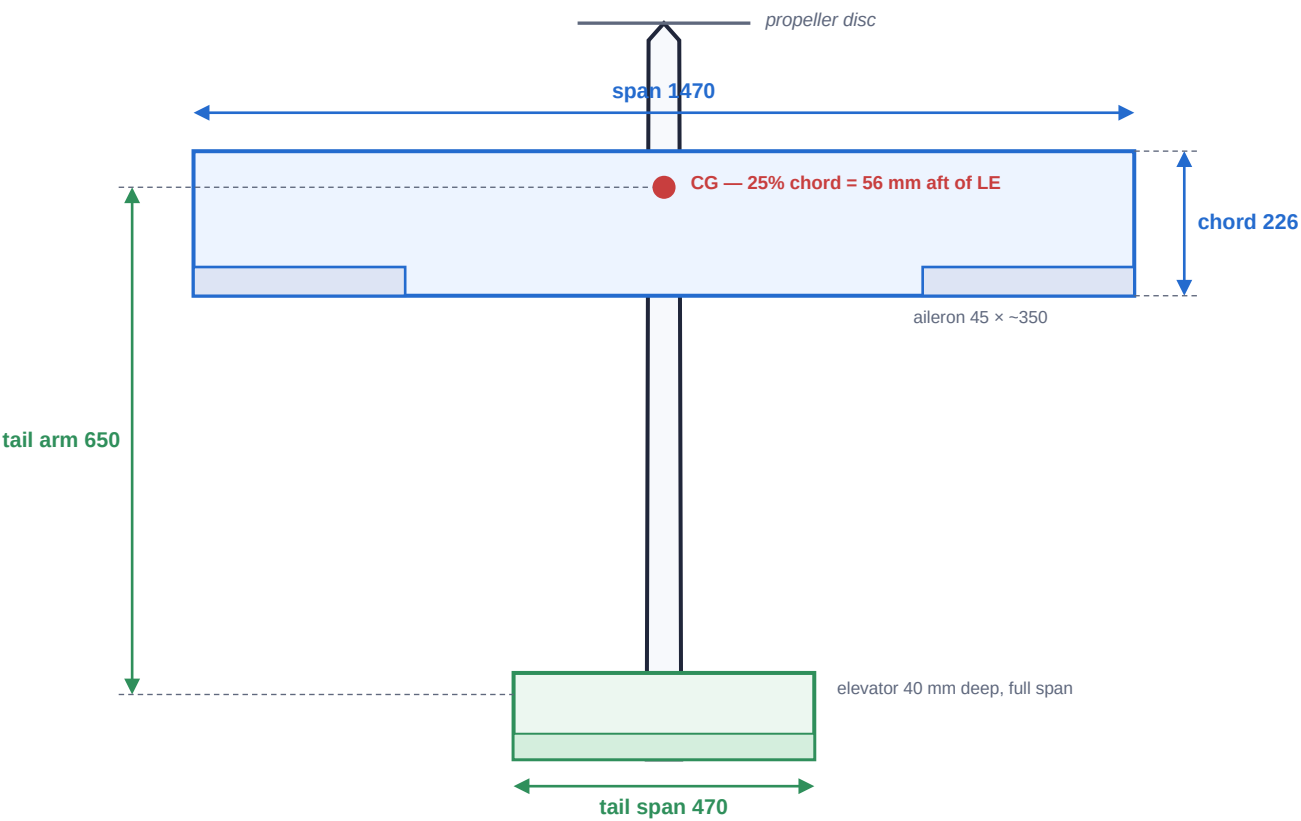
BUILD DECISION

- ESC: at least about 30 A for a measured 20–25 A setup, with adequate cooling and BEC capacity.
- Maiden CG: conservative 10–15% static margin; confirm the physical balance with the flight battery installed.
- Low rates: aileron/elevator 10–15°, rudder 20–25°. Confirm correct directions from behind the airplane.
- Have an experienced pilot perform or assist with the first flight; climb to height before trimming or testing stall behavior.

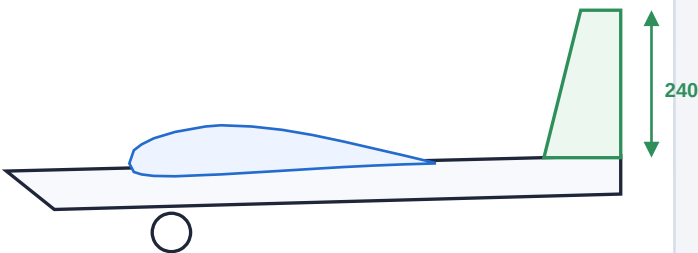
Step 5: the finished trainer — dimensioned three-view

The numbers from steps 1–4 turned into geometry you can build from. All dimensions in millimetres.

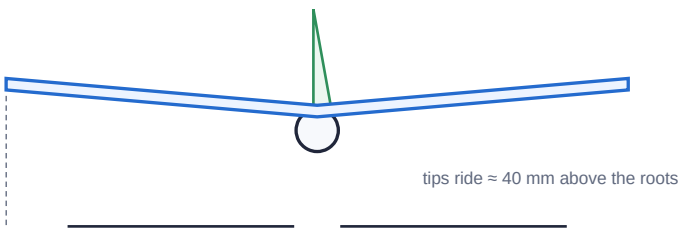
TOP VIEW — scale 1 : 300



SIDE VIEW — fin 240 tall, 125 chord



FRONT VIEW — dihedral $\approx 3^\circ$ per panel



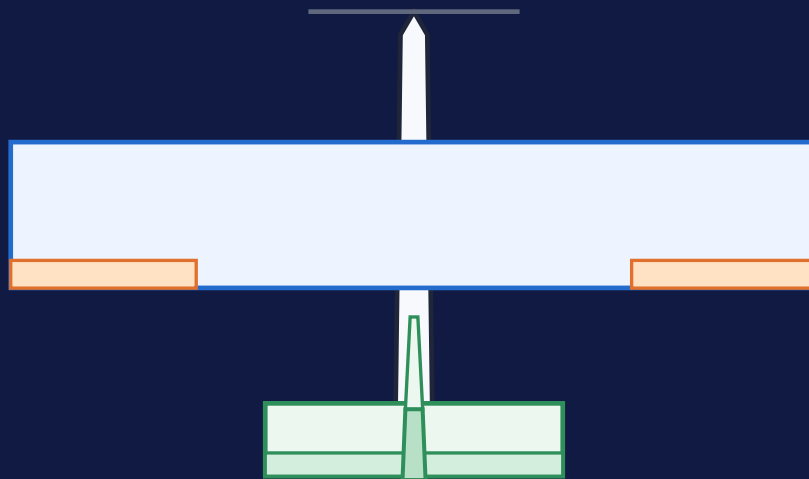
YOUR TURN

Redraw this three-view at 1:5 on paper, adding the rudder outline (35–40% of fin chord). Does the finished aircraft fit your building board and your car?

02

Understand airfoils

The wing section shapes the pressure field, lift range, drag, stall behavior, construction method, and usable speed range.



Think → calculate → build straight → test carefully → improve one change at a time.

How designers choose an airfoil

Nobody asks “which airfoil is best?” — they ask which airfoil fits the mission, the mass, and the Reynolds number.

THE QUESTIONS DESIGNERS ASK

- 1 How heavy is the airplane?**
Sets the lift the wing must make.
- 2 How fast should it cruise?**
Sets the working C_l and drag priorities.
- 3 How slowly must it land?**
Sets the C_{lmax} you need.
- 4 Will it fly inverted or do aerobatics?**
Pushes toward symmetric sections.
- 5 How easy must it be to build?**
Flat bottoms and gentle curves are friendlier.
- 6 How small is the wing chord?**
Sets Reynolds number — small chords need low-Re sections.

AIRFOIL ROLES AT A GLANCE

- **Clark Y**
trainers; easy flat-bottom build
- **NACA 2412**
sport all-rounder
- **S1223**
slow flight, heavy lifting
- **NACA 0012**
aerobatics, tails and fins
- **E387**
gliders, efficient cruise

Details: next five pages. Full vocabulary: reference pages 40–44.

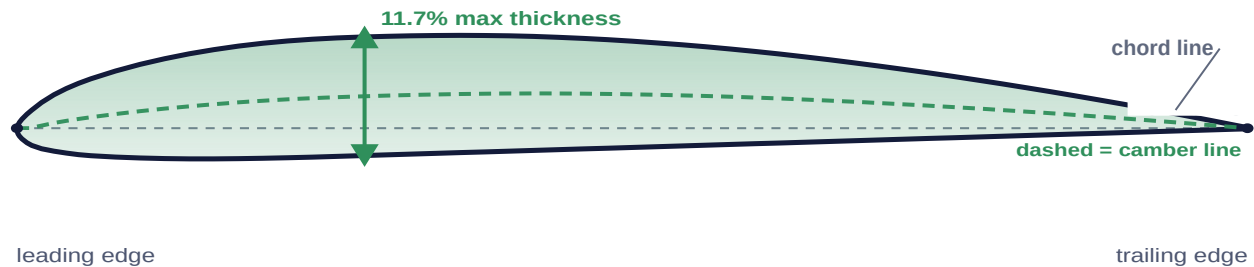
! One big idea: the wing pushes air down, so the air pushes the wing up (Newton’s third law). The airfoil decides how efficiently that trade happens, over what speed range, and how gently it fails at the stall. Choose it after mission, mass, and wing loading are known — never before.

HOW TO READ THE NEXT FIVE PAGES

Each page gives one family: its geometry (thickness, camber, and where they peak), what it is best at, why it flies the way it does, what to watch while building, and rough R/C tendencies scored 0–5. The scores are habits, not test data — wing loading and Reynolds number can overturn them.

Clark Y: the friendly builder's airfoil

A classic cambered section with a nearly straight aft lower surface



MAX THICKNESS

11.7%

near 28% chord

MAX CAMBER

3.4%

near 42% chord

PERSONALITY

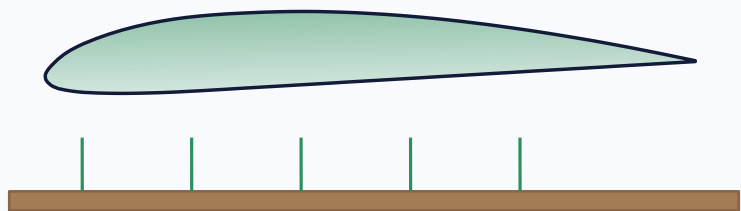
steady

forgiving, not especially fast

WHY IT FLIES THIS WAY

- 1 Moderate camber makes useful lift at a small angle of attack, so the airplane does not need to fly nose-high all the time.
- 2 The rounded leading edge and fairly thick nose give the airflow room to stay attached as the angle increases.
- 3 Its lower surface is close to straight over much of the rear half, which simplifies rib alignment during construction.

CONSTRUCTION PICTURE



A nearly straight aft lower surface makes alignment on a building board easier.

BEST FOR

- High-wing trainers
- Relaxed sport and scale models
- Medium-to-slow electric aircraft

BUILD NOTES

- Keep the leading edge smooth and truly rounded.
- Use the plan's jiggig or washout; avoid accidental wing twist.
- Leave enough spar depth near the thickest part.

WATCH-OUTS

- Not equally good inverted.
- Can become draggy when pushed fast.
- A thick airfoil cannot rescue excessive wing loading.

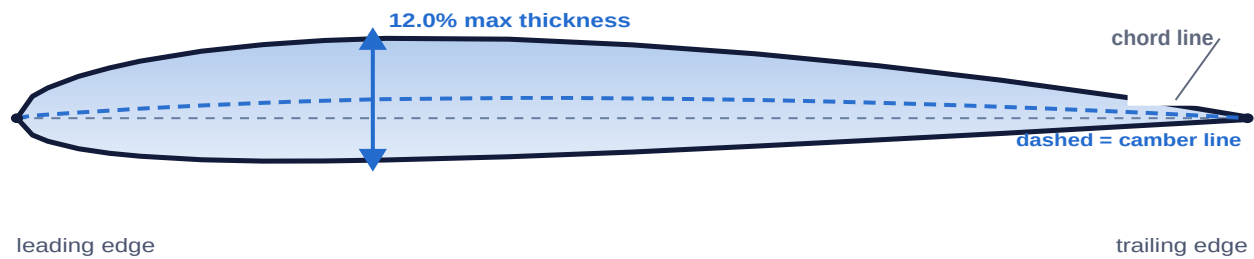
ROUGH R/C TENDENCIES (0-5)

Slow lift	■	■	■	■	■
Speed	■	■	■	■	■
Inverted	■	■	■	■	■

Not test data; wing loading and Reynolds number matter.

NACA 2412: the balanced all-rounder

A four-digit airfoil with moderate camber and useful structural thickness



MAX THICKNESS

12.0%

the "12" in its name

MAX CAMBER

2.0%

the first digit tells you this

CAMBER LOCATION

40%

the second digit tells you this

WHY IT FLIES THIS WAY

- 1 Two percent camber gives more upright lift than a symmetric section without creating an extreme undercambered shape.
- 2 Twelve percent thickness leaves useful room for a spar, servos, and a strong leading edge.
- 3 It is a compromise: reasonably efficient, reasonably gentle, and capable of sport flying without specializing in one job.

CRACK THE FOUR-DIGIT CODE

2

2% maximum camber

4

camber peak at 40% chord

12

12% maximum thickness

BEST FOR

- General sport models
- Scale aircraft
- Moderate aerobatics

BUILD NOTES

- The lower surface is curved, so use rib feet, tabs, or a jig.
- Measure the 40% camber location from the leading edge.
- Do not blunt the trailing edge more than the plan requires.

WATCH-OUTS

- Upright and inverted trim will differ.
- Full-size data may not match a small model's Reynolds number.
- Surface waves can erase a theoretical advantage.

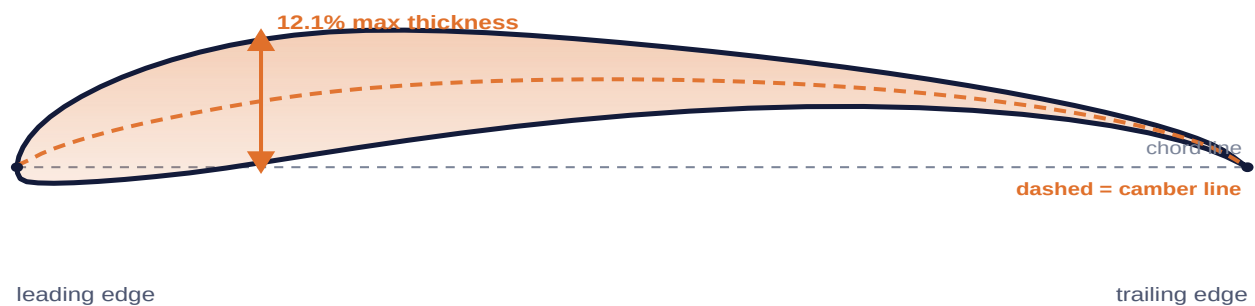
ROUGH R/C TENDENCIES (0-5)



Not test data; wing loading and Reynolds number matter.

Selig S1223: high lift at low model speeds

A strongly cambered, low-Reynolds-number section for slow flight and heavy lifting



MAX THICKNESS

12.1%

forward, near 20% chord

MAX CAMBER

8.7%

far more than the other examples

PERSONALITY

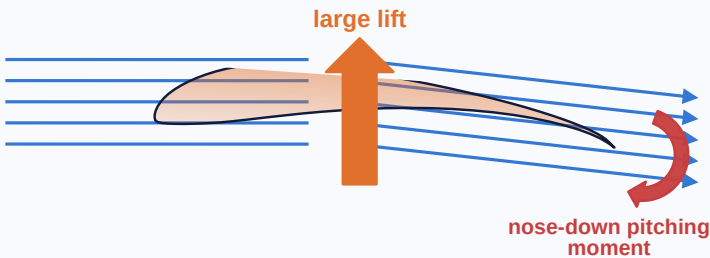
high lift

slow, powerful, draggy when fast

WHY IT FLIES THIS WAY

- 1 Very strong camber bends the airflow downward aggressively, allowing a large lift coefficient at low Reynolds numbers.
- 2 The concave lower surface helps the section's low-speed job but makes the rib and covering shape harder to preserve.
- 3 High camber usually brings a stronger nose-down pitching moment, so the tail and center of gravity must be designed to balance it.

HIGH LIFT COMES WITH A TRIM COST



The tail and center of gravity must balance that moment.

BEST FOR

- Slow flyers and STOL models
- Payload or lift demonstrations
- Models with short takeoff goals

BUILD NOTES

- Cut the undercamber accurately; a flat shortcut changes the airfoil.
- Support film covering so it does not bridge across the curve.
- Make the thin trailing edge stiff enough to stay straight.

WATCH-OUTS

- High drag at speed.
- Poor inverted performance.
- High lift does not mean stall-proof.

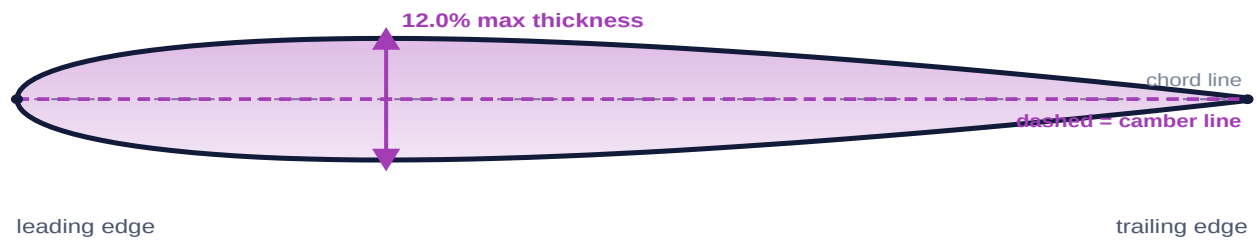
ROUGH R/C TENDENCIES (0-5)



Not test data; wing loading and Reynolds number matter.

NACA 0012: the same shape top and bottom

A 12%-thick symmetric section for aerobatics, tails, and predictable reversed flight



MAX THICKNESS

12.0%

the "12" in its name

MAX CAMBER

0%

the two leading zeros

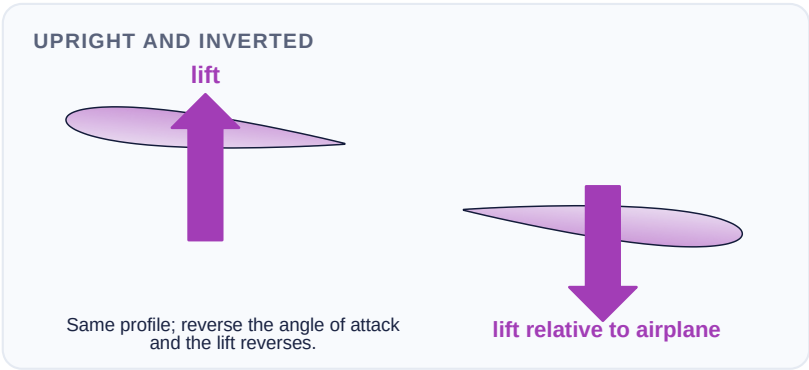
ZERO-ANGLE LIFT

about 0

for an ideal 2-D section

WHY IT FLIES THIS WAY

- 1 With no camber, the ideal section produces nearly zero lift at zero angle of attack.
- 2 A positive angle gives positive lift; the same negative angle gives nearly the same lift in the opposite direction.
- 3 The 12% thickness provides useful spar depth without favoring upright flight over inverted flight.



BEST FOR

- Aerobatic and pattern models
- Horizontal tails and fins
- Aircraft that fly inverted often

BUILD NOTES

- There is no flat bottom, so use jig tabs or a split-shell method.
- Mirror the ribs carefully; tiny asymmetry creates unwanted trim.
- Keep the leading edge radius equal on both sides.

WATCH-OUTS

- It still needs angle of attack to support weight.
- Symmetric does not automatically mean low drag.
- High wing loading still raises stall speed.

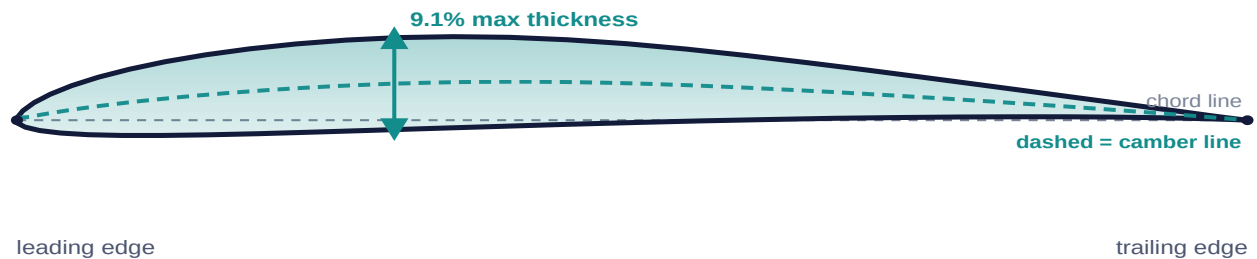
ROUGH R/C TENDENCIES (0-5)

Slow lift					
Speed					
Inverted					

Not test data; wing loading and Reynolds number matter.

Eppler airfoils: a toolbox, not one shape

E387 is a low-Re example; other Eppler numbers were designed for very different missions



E387 THICKNESS

9.1%

near 31% chord

E387 CAMBER

3.8%

near 41% chord

FAMILY RANGE

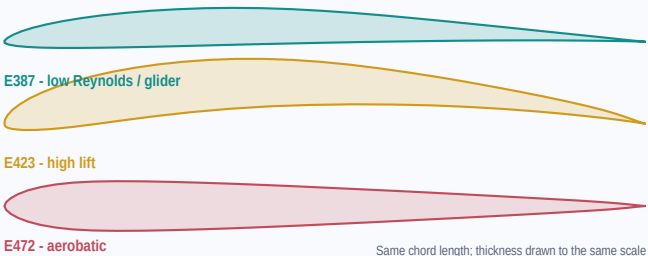
very wide

glider, high lift, aerobatic, and more

WHY IT FLIES THIS WAY

- 1 “Eppler” is a design family, not a single airfoil. The number after the name is essential.
- 2 E387 is relatively thin with moderate camber and is listed as a low-Reynolds-number profile, which suits efficient light models.
- 3 E423 is deeply cambered for high lift, while E472 is symmetric for aerobatics - proof that the family name alone predicts very little.

ONE NAME, VERY DIFFERENT JOBS



Always specify the number - “Eppler” by itself is incomplete.

BEST FOR

- E387: gliders and efficient models
- E423: specialized high lift
- E472: aerobatic applications

BUILD NOTES

- Choose the exact Eppler number for the mission.
- Accurate ribs and a smooth surface matter more on efficient sections.
- Match the airfoil to the model's expected Reynolds number.

WATCH-OUTS

- Do not specify only “Eppler.”
- Roughness and dents can spoil low-drag behavior.
- A profile tested at the wrong Reynolds number can mislead.

ROUGH R/C TENDENCIES (0-5)

E387 slow eff.

E387 speed

Build ease



Not test data; wing loading and Reynolds number matter.

The equations that connect shape to flight

Enough math to design intelligently — without pretending the model is simpler than it is

Core idea: in level flight $L \approx W$. Speed appears squared, so small speed changes matter a lot. C_L and C_D change with angle of attack, airfoil, Reynolds number, and surface finish.

DYNAMIC PRESSURE

$$q = \frac{1}{2} \rho V^2$$

q is the air's "push strength." ρ is air density and V is speed.

Double speed → four times q .

LIFT

$$L = q S C_L$$

S is wing area. C_L packages the effects of shape and angle of attack.

Level flight needs L about equal to weight.

DRAG

$$D = q S C_D$$

C_D is the drag coefficient. It rises when the wing is rough, badly shaped, or near stall.

More lift usually costs some drag.

STALL SPEED

$$V_{\text{stall}} = \sqrt{(2W / \rho S C_{L_{\text{max}}})}$$

W is weight and $C_{L_{\text{max}}}$ is the largest lift coefficient before stall.

Lighter + larger wing + higher $C_{L_{\text{max}}}$ = slower stall.

REYNOLDS NUMBER

$$Re = \rho V c / \mu$$

c is chord and μ is air viscosity. Re tells you how "sticky" the airflow acts.

Small, slow R/C wings usually have low Re .

WING GEOMETRY

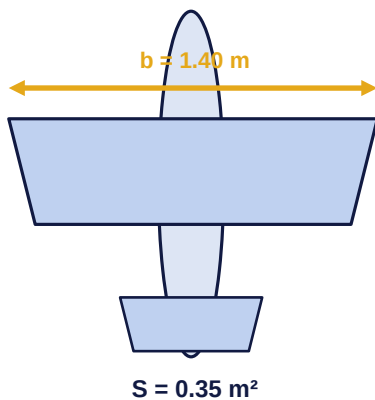
$$AR = b^2 / S \quad \text{and} \quad W/S$$

AR is aspect ratio; b is span. W/S is wing loading.

Long wings reduce induced drag; low wing loading helps slow flight.

WORKED EXAMPLE: A 1.5 KG SPORT MODEL

Given: wing area $S = 0.35 \text{ m}^2$, average chord $c = 0.25 \text{ m}$, span $b = 1.40 \text{ m}$, speed $V = 12 \text{ m/s}$, $C_L = 0.60$, $C_{L_{\text{max}}} = 1.20$. Use $\rho = 1.225 \text{ kg/m}^3$ and $\mu = 1.81 \times 10^{-5} \text{ Pa}\cdot\text{s}$.



- | | |
|--------------------|--|
| 1. Weight | $W = mg = 1.5 \times 9.81 = 14.7 \text{ N}$ |
| 2. Air push | $q = \frac{1}{2} \times 1.225 \times 12^2 = 88 \text{ Pa}$ |
| 3. Lift | $L = 88 \times 0.35 \times 0.60 = 18.5 \text{ N}$ |
| 4. Stall speed | $V_{\text{stall}} \approx 7.6 \text{ m/s} = 27 \text{ km/h}$ |
| 5. Reynolds number | $Re \approx 2.0 \times 10^5$ |
| 6. Aspect ratio | $AR = 1.40^2 / 0.35 = 5.6$ |

18.5 N of lift is greater than 14.7 N of weight, so at that speed and C_L the model would climb - or it could lower its angle of attack.

Where the equations come from

Nothing on the previous page is memorized magic — each result is a few lines of algebra away from $L = W$.

STALL SPEED, DERIVED

- 1 Level flight: $L = W$**
The wing exactly supports the weight — the starting point for everything.
- 2 $W = \frac{1}{2} \rho V^2 S C_L$**
Substitute the lift equation for L .
- 3 Slowest flight $\rightarrow C_L = C_{Lmax}$**
Flying slower means more lift coefficient; the most you can ask for is C_{Lmax} .
- 4 V^2 , isolated: $V^2 = 2W / (\rho S C_{Lmax})$**
Multiply both sides by 2, divide by $\rho S C_{Lmax}$.
- 5 Take the square root**
Speed must be positive — done.

$$V_{stall} = \sqrt{(2W / \rho S C_{Lmax})}$$

LOAD FACTOR IN A BANK, DERIVED

- 1 Bank angle ϕ tilts the lift**
Lift stays perpendicular to the wings, so it leans with them.
- 2 Vertical part: $L \cos \phi$**
Only this component fights gravity (see page 8).
- 3 Steady height: $L \cos \phi = W$**
Not climbing, not sinking — vertical forces balance.
- 4 $n \equiv L / W = 1 / \cos \phi$**
Define load factor n as lift over weight; divide step 3 by $W \cos \phi$.
- 5 $V_{stall} \propto \sqrt{L} \rightarrow$ multiply by $\sqrt{n}$**
Stall speed scales with the square root of the lift demanded.

$$V_{s,turn} = V_s \sqrt{n} = V_s / \sqrt{(\cos \phi)}$$



When you can re-derive a formula, you own it: you know exactly which assumptions it hides (steady flight, constant density, rigid wing) and when it stops being true.



YOUR TURN

Derive the general level-flight speed for any C_l (not just C_{Lmax}). Then compute the worked trainer's speed at $C_l = 0.4$ with $W = 14.7$ N, $S = 0.333 \text{ m}^2$. Check: you should get $\approx 13.4 \text{ m/s}$.

Induced drag: the price of making lift

The one equation missing from most beginner guides — and the reason gliders have long, skinny wings.

THE EQUATION

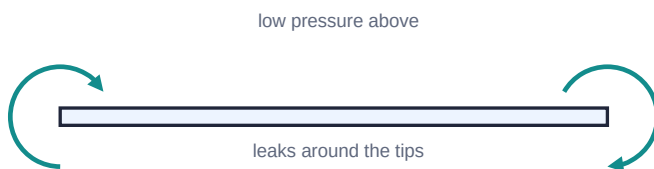
$$C_{D,i} = C_L^2 / (\pi \cdot AR \cdot e)$$

e is the span-efficiency factor: 1.0 for a perfect elliptical lift distribution, typically 0.7–0.85 for straight R/C wings. Everything else you already know.

READ IT LIKE AN ENGINEER

- C_L is squared: fly twice the lift coefficient, pay four times the induced drag.
- AR is in the denominator: longer wings pay less for the same lift.
- It has nothing to do with surface finish — this drag exists because lift exists.

WHERE IT COMES FROM



Air spills around each tip → trailing vortices → the wing flies in its own downwash → the lift vector tilts back. That backward tilt is induced drag.

WORKED NUMBERS — YOUR TRAINER

$AR = 6.5$, $e \approx 0.8$, so $\pi \cdot AR \cdot e \approx 16.3$

Cruise, $C_L = 0.60$:

$$C_{D,i} = 0.36 / 16.3 \approx 0.022$$

Slow pass, $C_L = 1.10$:

$$C_{D,i} = 1.21 / 16.3 \approx 0.074 \text{ — } 3.4\times \text{ more}$$

Slow flight is draggy flight: throttle up before you slow down.



Every gram costs twice: more weight → more C_L needed at the same speed → more induced drag → more power → bigger battery → more weight. This is why the mass budget on the worked-trainer pages matters so much.

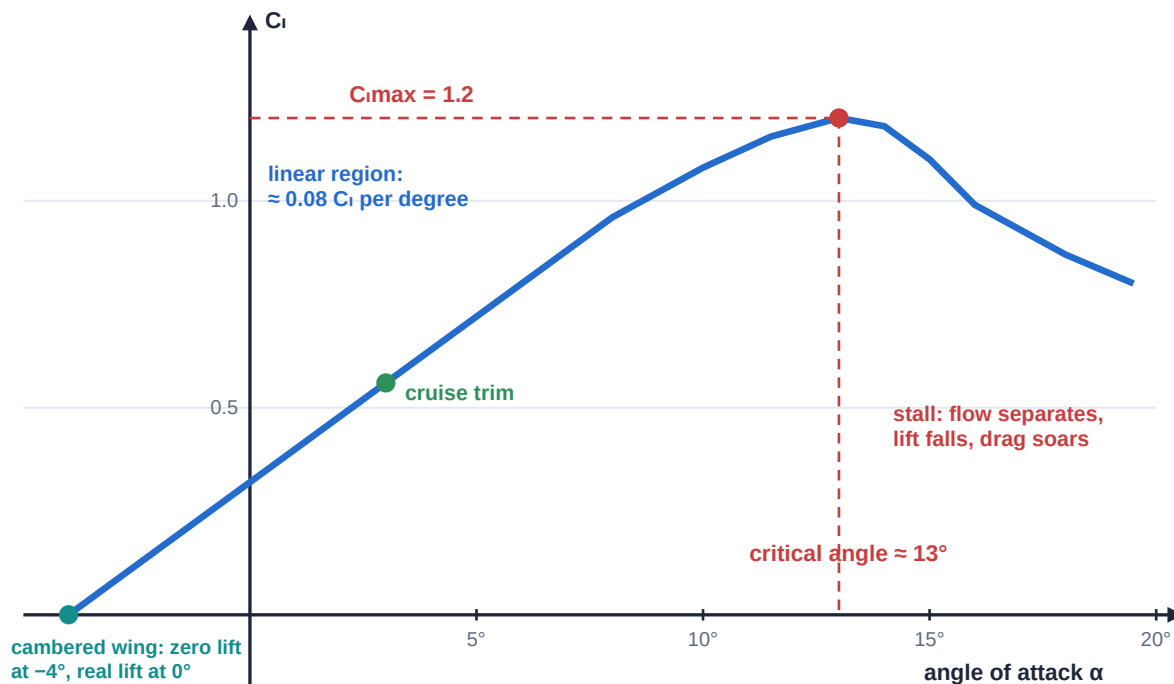


YOUR TURN

Halve the aspect ratio to 3.25 (same area). What happens to $C_{D,i}$ at $C_L = 0.6$? Now explain why an albatross — which soars for hours without flapping — has $AR \approx 15$.

The lift curve: what C_l actually does

One graph ties together angle of attack, C_{lmax} , and the stall — read it once and pages 14, 33, and 41 snap into place.



READ IT LIKE THIS

More angle $\rightarrow$ more lift, click for click, until the boundary layer gives up. The curve is honest: nothing warns you at 12.9° .

WHY IT MATTERS

The elevator commands α , so it commands C_l . A stall is exceeding the critical angle — possible at any airspeed, upright or inverted (page 41).

CAMBER SHIFTS IT

A symmetric section (NACA 0012) passes through the origin. Camber slides the whole line left — free lift at zero angle, at the cost of a nose-down pitching moment.

YOUR TURN

?

Sketch the NACA 0012 curve on top of this one (same slope, through the origin, $C_{lmax} \approx 1.0$ at model Reynolds numbers). Where do the two wings trim differently? Which stalls first at the same α ?

Reynolds number: why small models feel different air

Re appears on half the pages of this manual — here is the number line that makes it concrete.

THE DEFINITION

$$Re = \rho V c / \mu$$

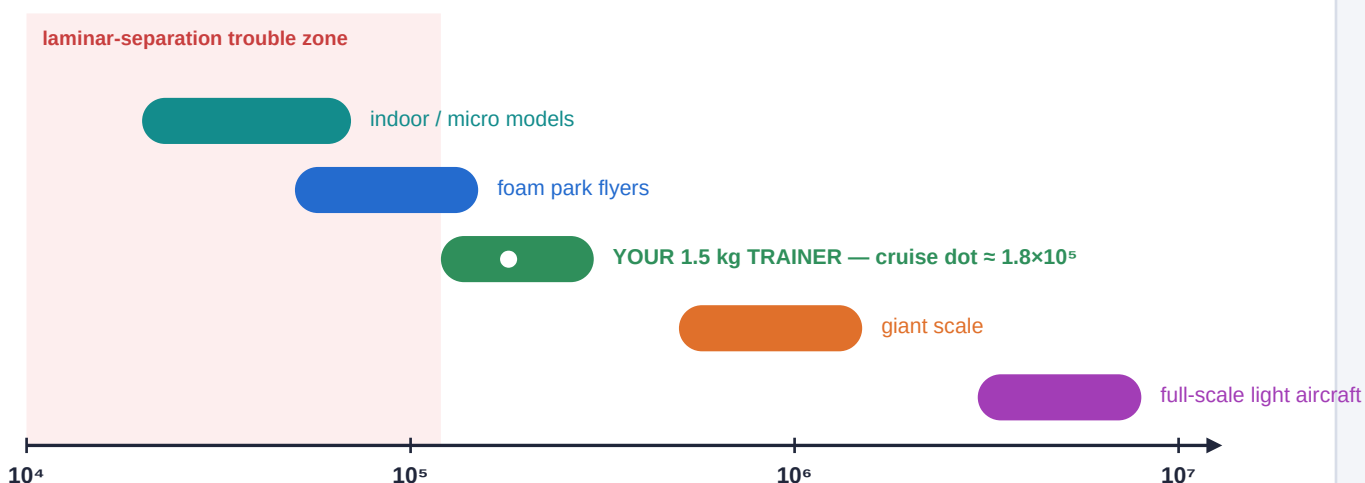
Speed $\times$ chord $\times$ density, against viscosity: how much the air behaves like a crowd in motion versus honey.

SEA-LEVEL SHORTCUT

$$Re \approx 68\,000 \times V \times c$$

V in m/s, c in metres. The trainer at 12 m/s with 0.226 m chord:
 $Re \approx 68\,000 \times 12 \times 0.226 \approx 1.8 \times 10^5$.

WHERE R/C AIRCRAFT LIVE ON THE Re SCALE



BELOW $\sim 10^5$, THINGS GET WEIRD

The boundary layer stays laminar, separates early, and may not reattach: C_{lmax} drops, drag rises, and stalls sharpen. Full-size airfoil data quietly stops being true.

WHAT TO DO ABOUT IT

- Prefer sections designed for low Re (E387, S1223 class) and data from UIUC low-speed tests.
- Keep the leading edge accurate and the skin smooth.
- Remember the tips: smaller chord = lower Re = they misbehave first.

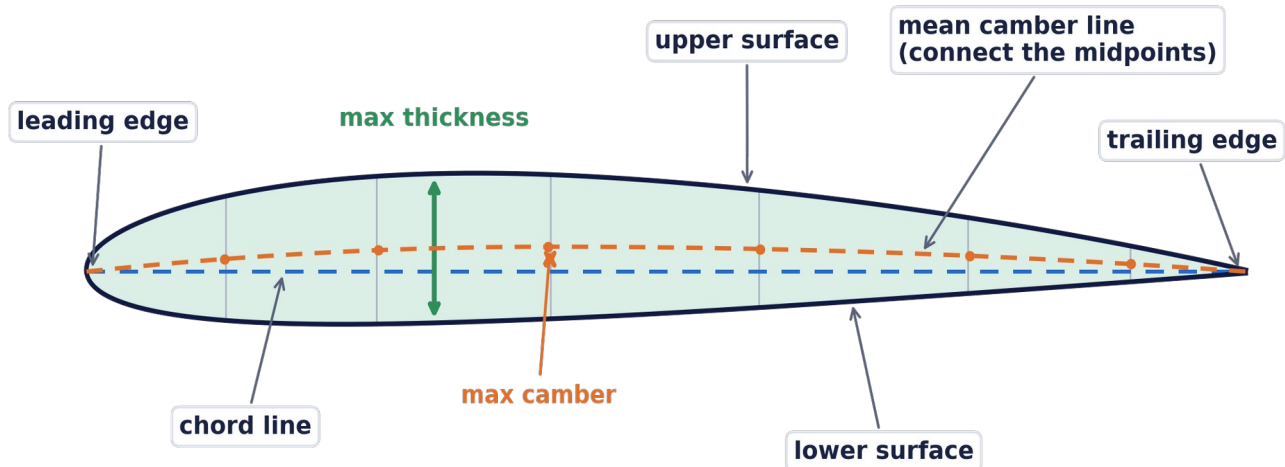
YOUR TURN

?

Compute Re at the trainer's wing in a slow landing: $V = 6$ m/s, $c = 0.226$ m. Is it inside the trouble zone? ($Re \approx 92\,000$ — marginal, which is why landings expose bad wings.)

Airfoil anatomy: camber, thickness, and chord

Camber, thickness, chord, and surfaces describe different parts of the same cross-section



At each chord position, the orange dot is halfway between the upper and lower surfaces. Connecting those dots makes the mean camber line.

CAMBER

The curve of the mean camber line - not the curve of the top or bottom by itself. Because the mean line is built from midpoints, camber depends on both surfaces. Max camber is usually written as a percentage of chord.

THICKNESS

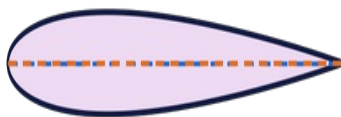
The distance between the upper and lower surfaces. Maximum thickness tells you the thickest part of the section, often as a percentage of chord. A thick airfoil can leave more room for a strong spar.

CHORD

The straight line from the leading edge to the trailing edge. Chord length is written c . Locations such as "40% chord" are measured back from the leading edge along this line.

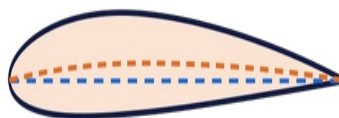
CURVED DOES NOT AUTOMATICALLY MEAN CAMBERED

Symmetric section



Curved surfaces, but a straight mean line

Cambered section



The mean line bends away from the chord

THE IMPORTANT TEST

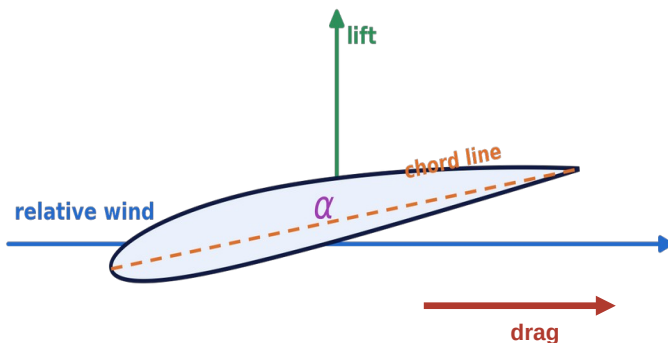
Ask what the mean camber line does.

- Straight mean line: zero camber, even when both surfaces are curved.
- Bent mean line: cambered airfoil.
- The largest gap between the mean line and chord is the maximum camber.

This is why a symmetric NACA 0012 has curved surfaces but 0% camber.

Angle of attack, airflow, lift, drag, and stall

These words describe the wing relative to the air - not just relative to the ground or horizon



angle of attack = angle between these two lines

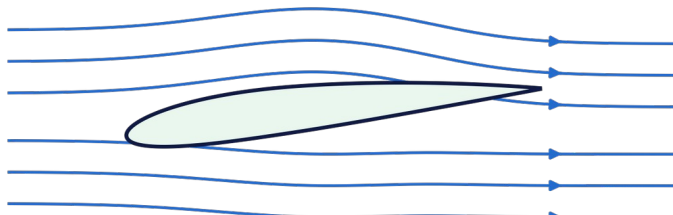
Lift is perpendicular to the relative wind. Drag points along it — downstream.

FOUR LINES AND ANGLES TO KEEP SEPARATE

- W Relative wind**
The airflow direction the wing feels. It points opposite the aircraft's path through the air.
- C Chord line**
A line fixed in the wing from leading edge to trailing edge.
- α Angle of attack**
The angle between the chord line and relative wind. It changes during flight.
- P Pitch attitude**
The nose angle relative to the horizon. It is not automatically the same as angle of attack.

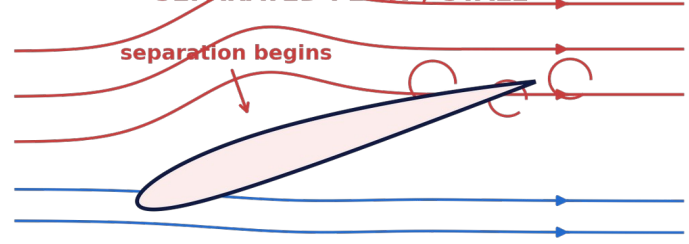
ATTACHED FLOW VERSUS SEPARATED FLOW

ATTACHED FLOW



The airflow follows the surface and leaves in an orderly way.

SEPARATED FLOW / STALL



A large wake forms. Lift stops increasing normally and drag rises sharply.

BOUNDARY LAYER

A very thin layer of air next to the surface. Skin friction slows it down. Even though the main airflow is fast, the air touching the wing has nearly zero speed relative to the surface.

SEPARATION

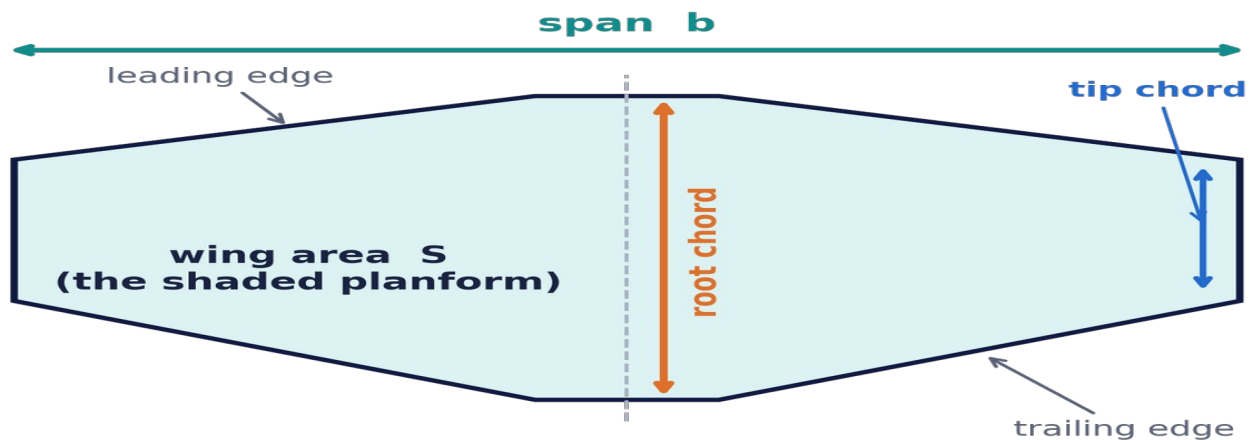
When the boundary layer no longer has enough energy to follow the surface, it peels away. The separated region creates a larger wake, more drag, and less predictable lift.

STALL

A stall happens when the wing exceeds its critical angle of attack and large-scale separation develops. It can happen at any airspeed. Low speed matters because holding the airplane up then requires more angle of attack.

Wing geometry: span, area, loading, and twist

An airfoil is one slice; the complete wing adds planform shape, spanwise angles, and structural limits



Planform means the outline seen from above. Wing area S is the area inside that outline.

SPAN, CHORD, AND AREA

Span b is tip-to-tip distance. Chord c is leading-edge to trailing-edge distance. On a tapered wing the chord changes, so designers may use a representative mean aerodynamic chord. Wing area S is the planform area.

ASPECT RATIO AR

$AR = b^2/S$. A long, narrow wing has a higher aspect ratio than a short, wide wing of the same area. Higher AR usually reduces induced drag, but it also increases bending loads and may need a stronger spar.

WING LOADING W/S

Weight divided by wing area. Lower wing loading usually means slower takeoff, landing, and stall speeds. Higher loading often feels faster and more solid in wind, but demands more speed and stronger structure.

ANGLES BUILT INTO THE COMPLETE WING

DIHEDRAL



Tips sit higher than the root. This adds roll stability.

INCIDENCE



The wing chord has a fixed angle to a fuselage reference line.

WASHOUT

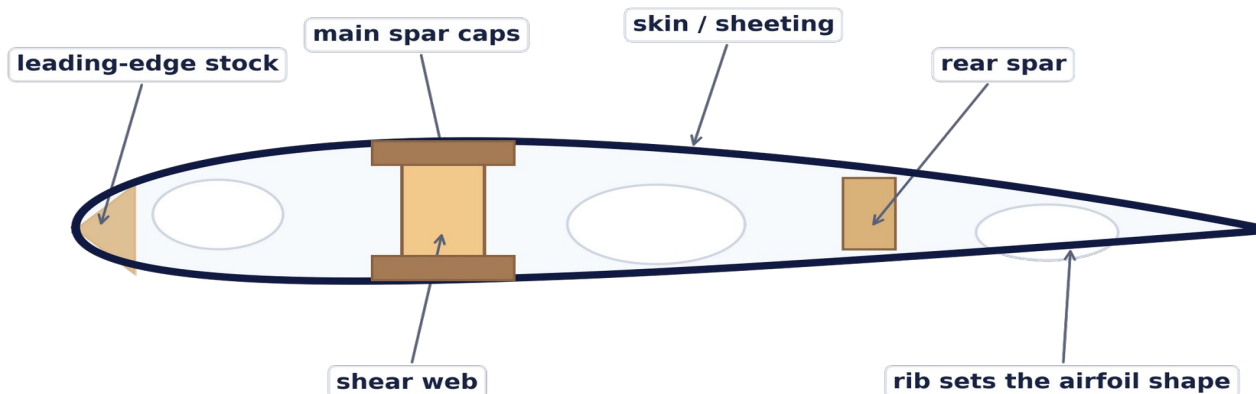


The tip has less incidence, helping the root stall before the tip.

Incidence is fixed by construction. Angle of attack changes in flight. Washout is an intentional change in incidence from root to tip.

Built-up wing terms: what the parts do

A good airfoil drawing only becomes a good wing when the structure holds that shape accurately



This is a conventional balsa-style cutaway. Foam, composite, and printed wings use different parts but must carry the same loads.

RIB, SPAR, AND SHEAR WEB

A rib is the cross-section frame that sets the airfoil shape. The spar is the main spanwise beam that resists bending. Spar caps carry most of the tension and compression; the shear web joins them and resists sliding forces.

SKIN, SHEETING, AND SURFACE FINISH

The skin creates the smooth outer surface. Sheeting can also form a torsion-resistant box with the spar. Dents, wrinkles, open gaps, and a blunt trailing edge disturb the boundary layer - especially on small, low-Reynolds-number models.

JIG, BUILDING BOARD, AND TWIST

A jig or flat building board holds parts while glue cures. Twist means the local wing angle changes along the span. Planned washout is useful; accidental twist can make one wing stall first and cause a sudden roll.

CENTER OF GRAVITY AND PITCHING MOMENT

The center of gravity (CG) is the balance point of the complete aircraft. Its location strongly affects stability. Pitching moment is the airfoil's tendency to rotate nose-up or nose-down; cambered sections often have a nose-down tendency that the tail must balance.

Equation symbol key: how to read the math page

A symbol is just a short name. The units tell you what kind of quantity it represents

Coefficient means a unitless score that packages the effects of shape and conditions. **Subscript max** means the largest value reached before the stated limit.

FORCES AND MOTION

- L** **Lift [N]**
Force perpendicular to the relative wind.
- D** **Drag [N]**
Force parallel and opposite to the relative wind.
- W** **Weight [N]**
Gravity force. $W = \text{mass} \times 9.81$ near Earth.
- V** **Airspeed [m/s]**
Speed through the air, not necessarily ground speed.

AIR AND PRESSURE

- ρ** **rho: density [kg/m³]**
How much air mass is packed into a volume.
- μ** **mu: viscosity [Pa s]**
How strongly the air resists layers sliding past each other.
- q** **Dynamic pressure [Pa]**
 $q = 1/2 \rho V^2$. A measure of the airflow's push strength.
- α** **alpha: angle of attack [deg]**
The angle between chord line and relative wind.

WING GEOMETRY

- S** **Wing area [m²]**
Planform area of the complete wing.
- b** **Span [m]**
Tip-to-tip distance.
- c** **Chord [m]**
Leading-edge to trailing-edge distance.
- AR** **Aspect ratio [none]**
 $AR = b^2/S$. A shape ratio, so it has no units.

COEFFICIENTS AND RESULTS

- C_L** **Lift coefficient, C_L**
How strongly the wing makes lift at this angle and condition.
- C_D** **Drag coefficient, C_D**
How strongly the aircraft makes aerodynamic drag.
- C_{Lmax}** **Maximum lift coefficient, C_{Lmax}**
The greatest C_L reached before lift begins to fall near stall.
- Re** **Reynolds number [none]**
Compares motion effects with viscous or "sticky" effects.

HOW THE MAIN EQUATIONS CONNECT

FORCE

$$L = q S C_L \quad D = q S C_D$$

Air push x area x coefficient

SCALING

$$Re = \rho V c / \mu \quad AR = b^2/S$$

Size, speed, air, and planform

STALL SPEED

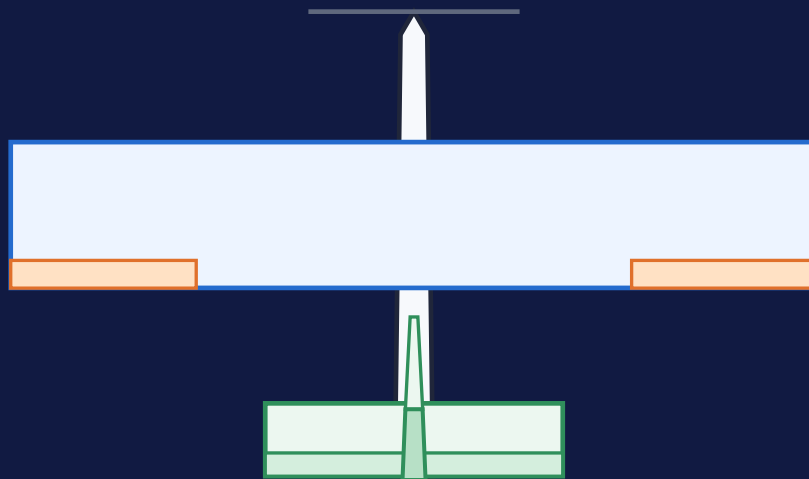
$$V_{\text{stall}} = \sqrt{2W / \rho S C_{L\text{max}}}$$

More weight raises it; more area lowers it

03

Build the structure

The aerodynamic design only exists in the air if the finished model is straight, stiff enough, light enough, and reliably connected.



Think → calculate → build straight → test carefully → improve one change at a time.

Choose a construction method that matches the mission

The “best” material is the one that meets stiffness, mass, repair, cost, and accuracy needs.

BALSA / PLYWOOD

- Light, stiff, repairable, and precise.
- Excellent for built-up wings and box fuselages.
- Requires grain awareness, jigs, and finishing.

FOAMBOARD / DEPRON

- Fast and inexpensive prototypes.
- Folded skins can form strong box structures.
- Paper and hot glue can add surprising mass.

EPP / EPO / MOLDED FOAM

- Tough and practical for trainers.
- Adhesive compatibility matters.
- Local reinforcement is needed at mounts and hinges.

3D PRINT / COMPOSITE / HYBRID

- Can create complex, repeatable shapes.
- Printed shells can be heavy or brittle.
- Composites are efficient but require process control and safety gear.



Prototype the hardest joint first: wing joiner, landing-gear plate, motor mount, removable wing, or control hinge. A weak joint can erase the advantage of strong material.

Adhesives: match the glue to the load and material

Glue is part of the structure. More glue is often heavier—not stronger.

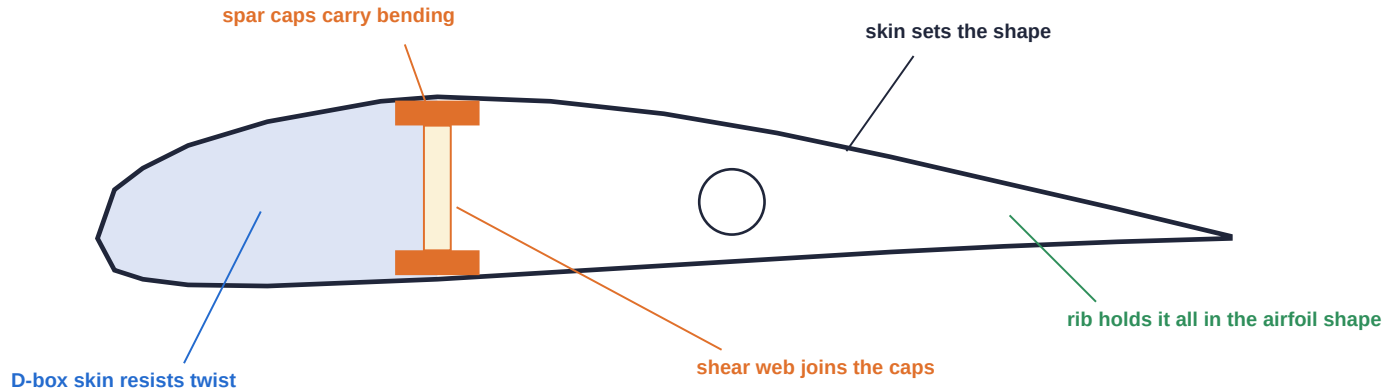
ADHESIVE	GOOD USE	ADVANTAGE	WATCH OUT
Thin CA	Tight balsa joints	Fast, light wicking bond	Brittle; fumes; bonds skin instantly
Medium / thick CA	Small gaps, field repairs	Fast and convenient	Use foam-safe type where required
Aliphatic / PVA	Wood-to-wood joints	Sandable, tough, low odor	Needs clamping and cure time
Epoxy	Motor mounts, joiners, mixed materials	Gap filling; strong when joint is designed well	Heavy; mixing accuracy; sensitization risk
Hot glue	Foamboard prototypes	Fast; fills gaps	Heavy; softens with heat; bulky fillets
Contact / foam adhesive	Foam skins and large areas	Flexible; spreads load	Ventilation; compatibility; cure time

 Design the joint so parts fit and loads flow through overlapping material, doublers, tabs, or gussets. Glue should transfer load—not bridge a badly fitting gap by itself.

Wing structure: bending, shear, and torsion

A wing must resist bending upward and twisting under aerodynamic and control loads.

CROSS-SECTION THROUGH THE WING (one rib bay)



lightening hole: keeps the rib light where stresses are low

ROUGH ROOT MOMENT

$$M_{\text{root}} \approx n W b / 8$$

Idealized uniformly loaded straight wing. n is load factor, W weight, b span.

LOAD PATH

- Upper and lower spar caps carry most bending tension/compression.
- Shear web keeps the caps working together.
- Closed leading-edge skin or D-box resists twist.

CRITICAL LOCATIONS

- Wing root and joiner.
- Landing-gear attachment.
- Servo and control-surface cutouts.

DESIGN HABIT

- Avoid abrupt stiffness changes.
- Reinforce load entry points.
- Test a spare joint to failure.

YOUR TURN

? Your 1.5 kg trainer pulls a 60° banked turn ($n = 2$). Estimate the root moment: $M_{\text{root}} \approx 2 \times 14.7 \text{ N} \times 1.47 \text{ m} / 8 \approx 5.4 \text{ N}\cdot\text{m}$ — like holding a 1 kg water bottle half a metre from your wrist. Would a bare 6 mm balsa spar take it?

A repeatable wing-building sequence

Accuracy is easiest when the wing is built over a flat datum and checked before every irreversible step.

1

PREPARE

Full-size plan, protected flat board, matched ribs, straight spars, labeled left/right parts.

2

PIN THE DATUM

Lower spar or lower skin fixed straight; root and tip blocks establish incidence and washout.

3

BUILD THE SKELETON

Ribs square to the board; add spars, web, leading edge, trailing edge, servo mounts.

4

LOCK IN TORSION

Add D-box, top skin, or folded skin while the panel is still held in the jig.

5

JOIN PANELS

Set dihedral with a gauge; reinforce joiner and root; verify both incidences match.

6

FIT CONTROLS

Cut free surfaces, bevel, hinge, install horns, route leads, and check full travel.

7

FINISH AND RECHECK

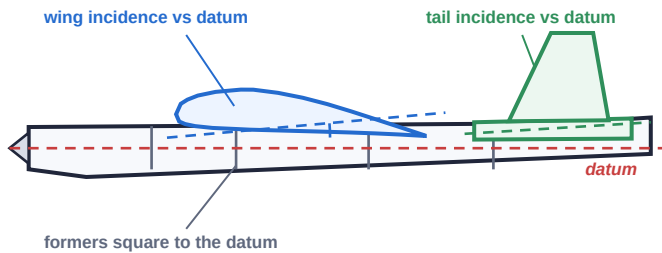
Cover symmetrically; remove any warp with heat and measurement—not guesswork.

! Do not “build the warp out later.” The covering stage can correct small errors, not a fundamentally crooked structure.

Fuselage and tail: make the reference geometry real

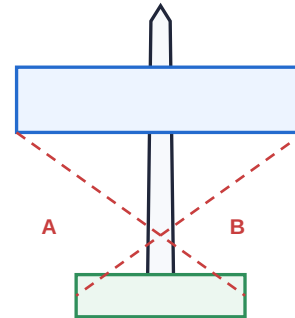
A straight fuselage and square tail reduce trim drag and prevent mysterious handling problems.

SIDE — every angle from ONE datum



Draw the datum on the plan and on the board. Wing and tail incidences are measured from it — never from each other, never by eye.

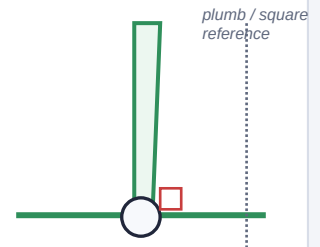
TOP — diagonals must match



A = B, or the tail is skewed

Tape from each wingtip to the opposite stab tip before the glue cures.

REAR — fin vertical



90° to the tailplane

Sight it from the front AND the rear — twist hides from one side.

FUSELAGE BOX

- Use a straight centerline and square formers.
- Add doublers where wing, gear, battery, and motor loads enter.
- Provide cooling inlet and a larger exit path.

TAIL MOUNTING

- Horizontal tail square to centerline and at intended incidence.
- Fin vertical when viewed from front and rear.
- Use temporary strings or diagonal measurements before gluing.

ACCESS

- Battery removable without disturbing controls.
- Receiver and linkages inspectable.
- Wing fasteners reachable.

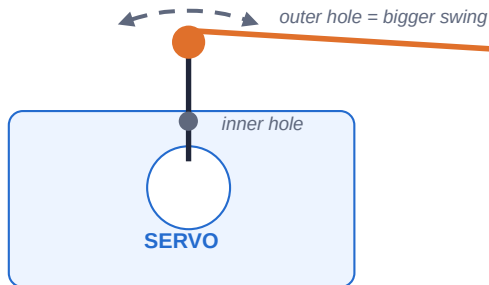
CRASHWORTHINESS

- Let inexpensive parts fail before expensive ones.
- Avoid battery contact with sharp structure.
- Secure canopy and hatch.

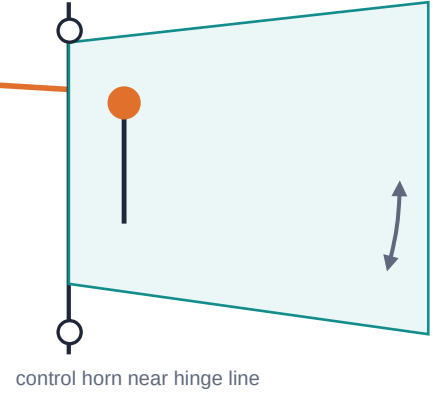
Hinges, horns, and pushrods

The control system must move freely with almost no looseness and remain secure under vibration.

inner servo hole = more force, less travel



surface swings through this arc



HINGE LINE

- Straight and aligned.
- Small gap.
- Enough bevel for travel.
- Every hinge bonded on both sides.

CONTROL HORN

- Load spread into solid structure.
- Hole close to hinge line.
- No soft foam crushing beneath it.

PUSHROD

- Short, straight, and supported.
- No buckling under push load.
- Thread lock only where appropriate.

BENCH TEST

- Full travel without binding.
- Surface returns to exact neutral.
- No buzz, flex, or lost motion.

Alignment, covering, and finish

A glossy finish cannot compensate for mismatched angles or a warped wing.

BEFORE COVERING

- ☐ Wing panels match in incidence and washout.
- ☐ Horizontal tail is square to the fuselage centerline.
- ☐ Fin is vertical.
- ☐ Control surfaces straight; hinge gaps consistent.

DURING COVERING

- ☐ Shrink opposite sides in stages.
- ☐ Support the panel on a flat jig.
- ☐ Recheck twist while the material is still warm.
- ☐ Keep heat away from foam and plastic parts.

AFTER COVERING

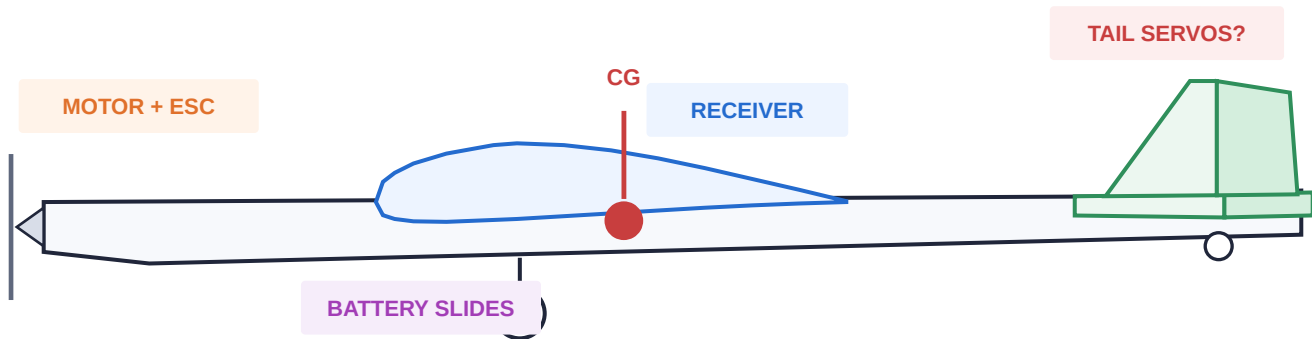
- ☐ Measure root and tip angles again.
- ☐ Check the wing join and fasteners.
- ☐ Verify all controls move freely.
- ☐ Open cooling paths and servo hatches.

FINISHING DISCIPLINE

- ☐ Paint and filler have real mass.
- ☐ Keep the tail especially light.
- ☐ Balance the spinner and propeller.
- ☐ Mark the CG points permanently under the wing.

Weight distribution and component placement

The lightest airplane is not enough; mass must also be located where the design can balance without dead ballast.



PLACEMENT ORDER

- Fix immovable masses first: motor, landing gear, servos.
- Create a useful battery adjustment range.
- Keep receiver and antennas away from high-current wiring.

AVOID DEAD BALLAST

- Move battery, servos, or equipment before adding lead.
- A heavy tail requires a much larger nose correction.
- Long tail pushrods may be lighter than tail-mounted servos.

MOMENT BALANCE

$$\Sigma(m \times \text{distance}) = 0$$

Measure distances from the desired CG. Nose moments balance tail moments.

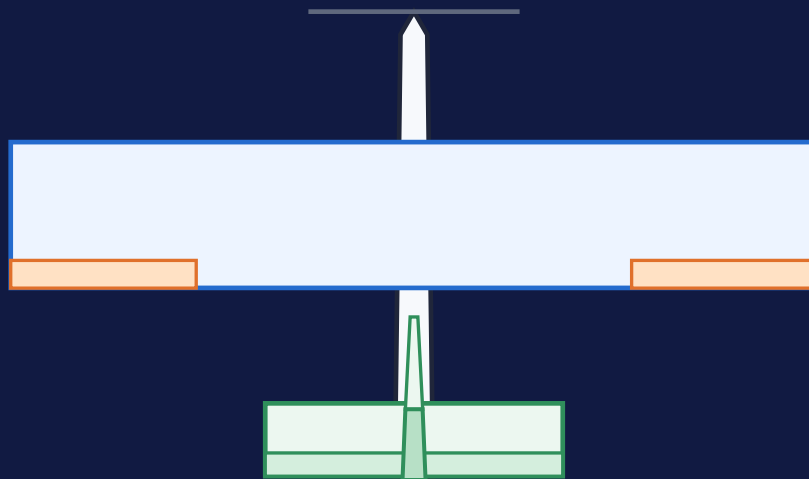
INSTALLATION RULE

- Secure the battery for all flight loads.
- Protect it from sharp edges.
- Keep access simple enough that it is always installed correctly.

04

Choose the systems

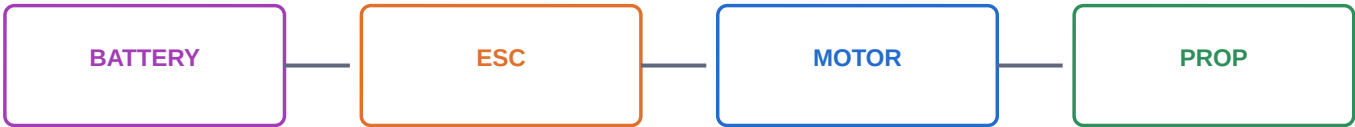
The power system, propeller, battery, radio, servos, and wiring must work as one tested combination—with thermal and electrical margin.



Think → calculate → build straight → test carefully → improve one change at a time.

Power, thrust, and propeller basics

Power tells you how quickly work can be done. Thrust is the force that accelerates the airplane.



Voltage sets potential speed. Current creates torque. The propeller turns power into thrust.

TRAINER

100–150 W/kg

broad starting range

SPORT

150–220 W/kg

broad starting range

HIGH PERFORMANCE

220–350+ W/kg

mission dependent

PROPELLER DIAMETER

- More disk area; usually more static thrust.
- Raises motor torque and current.
- Needs ground and fuselage clearance.

PROPELLER PITCH

- Higher pitch tends toward higher pitch speed.
- Can overload the motor.
- Prop efficiency depends on aircraft speed and operating point.

 Never choose the propeller from diameter and pitch alone. Use motor/prop/battery test data and verify voltage, current, power, temperature, and RPM with the actual installation.

Motor Kv, battery, ESC, and BEC

The labels describe different limits. No single number tells you whether the system is safe.

MOTOR KV

Kv is approximately no-load RPM per volt. It is not a power rating. Propeller load, winding resistance, voltage, cooling, and current limit determine usable output.

BATTERY

Cell count sets nominal voltage. Capacity sets stored charge. C rating claims current capability, but voltage sag, connector quality, age, and temperature also matter.

ESC

Must tolerate measured current, battery voltage, switching load, and temperature. Provide cooling and margin. Program brake and timing only when the motor/prop combination requires it.

BEC / RECEIVER POWER

The BEC powers receiver and servos. Confirm voltage compatibility and current capacity—especially with many digital servos or high-voltage servos.

ELECTRICAL POWER

$$P = V \times I$$

A 3S pack near 11.1 V delivering 20 A supplies about 222 W before losses.

IDEAL NO-LOAD RPM

$$\text{RPM} \approx \text{Kv} \times V$$

Loaded RPM is lower. This estimate does not approve a propeller.

 LiPo safety: use the correct charger mode and cell count, inspect for damage or swelling, charge in a suitable fire-resistant location, never short the pack, and follow the battery manufacturer's storage and disposal instructions.

Worked propulsion choice for the 1.5 kg trainer

Select a tested combination, then verify it with a wattmeter and temperature checks.

TARGET POWER

≈ 225 W

150 W/kg × 1.5 kg

NOMINAL VOLTAGE

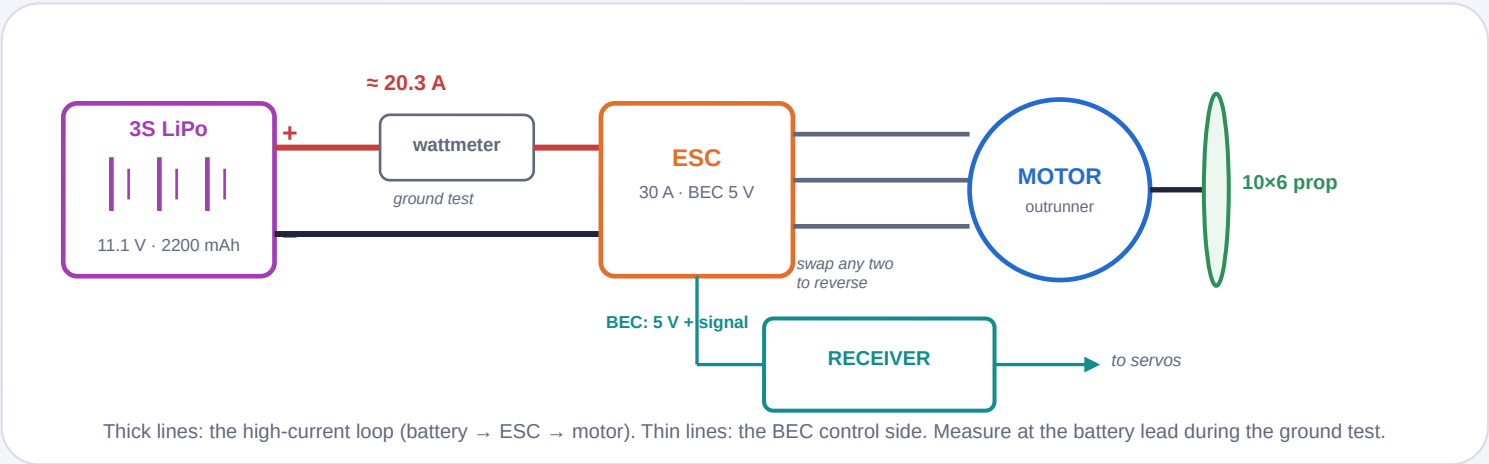
11.1 V

3S LiPo

IDEAL CURRENT

≈ 20.3 A

225 W ÷ 11.1 V



CANDIDATE SYSTEM

- 3S 2200 mAh LiPo.
- Outrunner and propeller combination documented around 220–240 W.
- ESC rated at least 30 A with an adequate BEC.
- Connector and wiring sized for measured current.

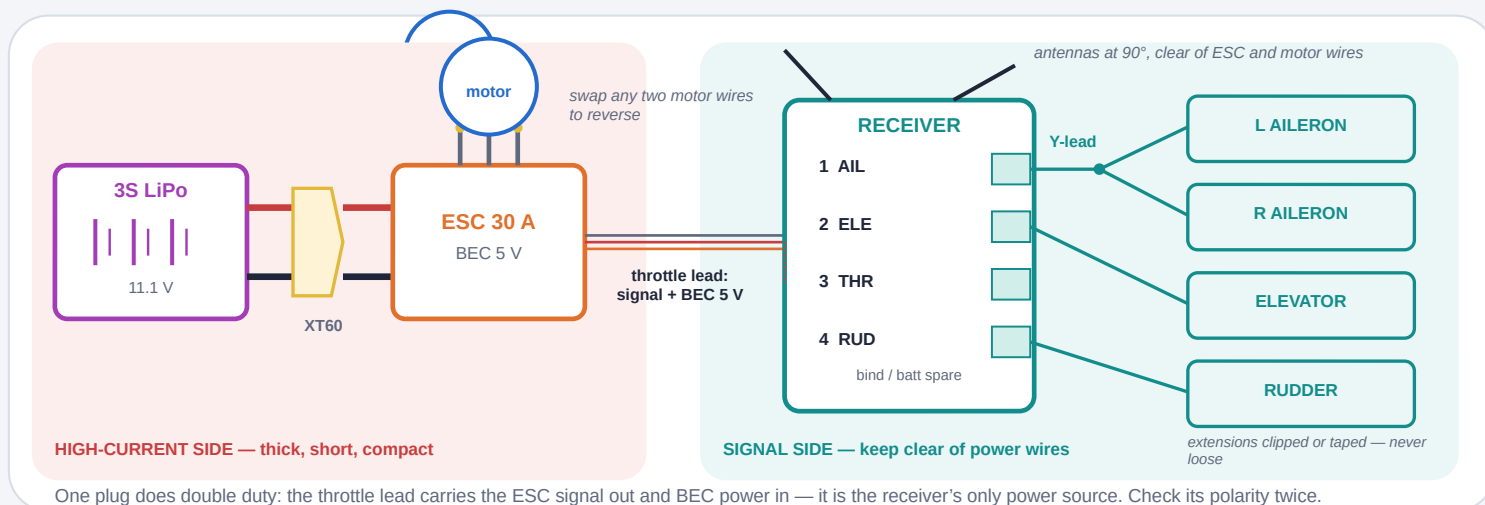
GROUND TEST

- Restrain aircraft safely; stay outside propeller arc.
- Measure current and power at full throttle briefly.
- Check motor, ESC, battery, and connectors for heat.
- Repeat with cowling installed to verify cooling.

Do not “prop up” until the current reaches the ESC label. Current limits are maximums, not performance targets. Leave thermal and operating margin.

Radio, servo, and wiring architecture

A clean installation makes failures less likely and troubleshooting much faster.



SERVO SELECTION

- Enough torque at the actual supply voltage.
- Speed appropriate to aircraft.
- Metal gears are not automatically better.
- Mount rigidly; isolate vibration where specified.

RECEIVER INSTALLATION

- Antennas oriented as manufacturer specifies.
- Away from motor wires, ESC, carbon shielding, and metal.
- Secure against vibration but do not crush.
- Easy bind and status access.

FAILSAFE

- Set throttle to safe/off position.
- Choose sensible control positions.
- Test by switching transmitter off only according to radio instructions.

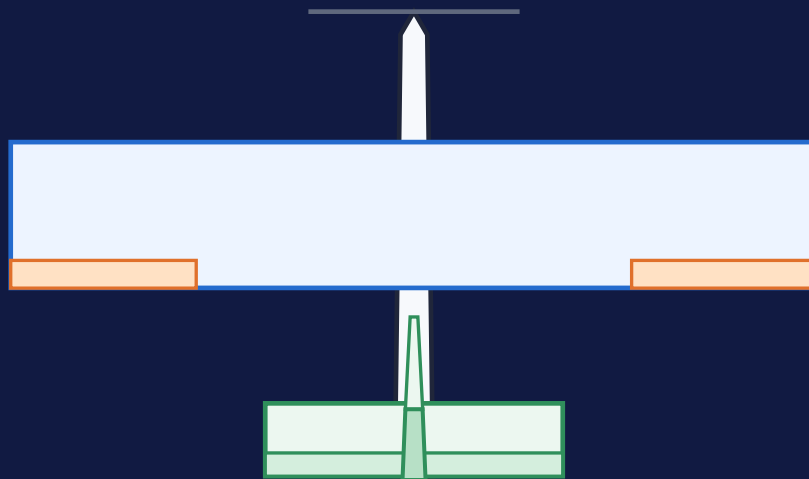
WIRE MANAGEMENT

- Strain relief at connectors.
- No rubbing on sharp edges.
- Servo extensions positively retained.
- High-current loops kept compact.

05

Test, fly, and improve

A maiden flight is a controlled experiment. The goal is not to prove the airplane perfect—it is to collect safe evidence.



Think → calculate → build straight → test carefully → improve one change at a time.

Preflight inspection: airframe, controls, power, and radio

Use a written sequence. Memory is weakest when excitement is highest.

AIRFRAME

- ☐ Wing and tail secure.
- ☐ No cracks, loose covering, or warps.
- ☐ Propeller undamaged and correctly fitted.
- ☐ Landing gear and wheels secure.

CONTROLS

- ☐ Correct direction viewed from behind.
- ☐ Full movement; no binding or buzz.
- ☐ Neutral positions match marks.
- ☐ Hinges, horns, clevises, and retainers secure.

POWER

- ☐ Correct battery and charge state.
- ☐ Pack restrained; hatch latched.
- ☐ Cooling path clear.
- ☐ Throttle cut works; prop area clear.

RADIO / RANGE

- ☐ Correct model memory.
- ☐ Low rates selected.
- ☐ Failsafe verified.
- ☐ Range test completed as the radio manufacturer specifies.



Same order every time: airframe → controls → power → radio. Anything that fails gets fixed before flying — not watched during it.

The maiden-flight plan

Decide what will happen before takeoff—including the abort plan.

1

CONDITIONS

Large familiar field, mild steady wind, good light, charged batteries, experienced helper.

2

GROUND CHECK

CG, directions, range, power, taxi/hand-launch plan, timer, and emergency throttle cut.

3

TAKEOFF

Use a firm smooth launch or straight takeoff. Do not climb at an extreme angle.

4

CLIMB TO HEIGHT

Reach a safe altitude before trimming, slowing, or making aggressive turns.

5

TRIM LIGHTLY

Correct one axis at a time. If large trim is needed, land and inspect instead of masking a problem.

6

BASIC CHECKS

Gentle turns, power-off glide, control response, and temperature. No low-altitude stall test.

7

LAND EARLY

Use a wide pattern and extra speed. Save advanced testing for later flights.



Abort if the airplane is badly out of trim, controls feel wrong, power is weak, or the structure sounds or behaves abnormally.

Trim and tune in a safe order

Do not change CG, incidences, control travel, and mixes all at once. You will not know what fixed—or caused—the behavior.

1. MECHANICAL ZERO

Straight surfaces, centered servo arms, equal link lengths. Use transmitter trim only for small flight corrections.

2. CG FIRST

Confirm balance and pitch behavior before adding mixes. Move aft only in small steps after stable flights.

3. THROWS + EXPO

Set maximum rate for required maneuvers; use expo to shape center feel, not hide slop.

4. AILERON DIFFERENTIAL

Adjust to reduce adverse yaw. Observe coordinated turns at safe height.

5. POWER EFFECTS

Correct thrust-line or mixing only after checking CG, alignment, and control neutral.

6. FLAPS / SPECIAL MODES

Add elevator compensation gradually at altitude; use slow deployment.

 Keep a flight log: configuration, battery, wind, CG, throws, trim clicks, temperatures, flight time, observations, and next single change.

Troubleshooting by symptom

Land first. Then separate aerodynamic, mechanical, electrical, and pilot-input causes.

SYMPTOM	POSSIBLE CAUSES	TEST	LIKELY ACTION
Pitches up with throttle	CG aft; thrust line; wing incidence; flexible tail	Check CG/alignment; static run; inspect flex	Correct cause; avoid large mix first
Needs much up elevator	CG forward; tail incidence; heavy model	Verify balance and neutral; glide behavior	Move mass / correct geometry carefully
Rolls one way hands-off	Wing warp; lateral balance; aileron neutral	Measure both panels; support at centerline	Remove warp; balance; reset linkage
Wanders in yaw	Small fin; linkage slop; aft CG; turbulence	Inspect rudder; compare tail volume	Repair linkage; CG forward; larger fin if proven
Tip drops near stall	Taper/warp; tip incidence; aileron input	Measure washout; inspect surface roughness	Correct twist; reduce taper effect; fly faster
Servo chatters / buzzes	Binding; overload; poor power; RF/noise	Disconnect linkage; load-test BEC	Fix geometry; power system; replace damaged servo
Short flight time	High current; damaged pack; drag; overprop	Wattmeter; capacity; temperatures	Use correct prop; reduce drag; replace pack

One-page preliminary design worksheet

Print this page or duplicate it in the PowerPoint. Fill every box before cutting material.

MISSION + FIELD

Primary mission: _____
Field / launch / landing limits: _____

one primary mission; numbers, not adjectives

MASS BUDGET

Target all-up mass: _____ g
Measured estimate: _____ g
Reserve: _____ %

keep a 10–15% reserve for what you forgot

WING

Target loading: _____ g/dm²
Area S: _____ dm²
AR: _____
Span b: _____ mm
Mean chord $\bar{c}$: _____ mm

$S = m \div \text{loading} \cdot b = \sqrt{(AR \cdot S)} \cdot c = S \div b$

STALL + SPEED

Assumed C_{lmax} : _____
Estimated V_s : _____ m/s
Approach target: _____ m/s

$V_s = \sqrt{(2W / \rho S C_{lmax})} \cdot \text{approach} \approx 1.3 V_s$

HORIZONTAL TAIL

V_h : _____
Tail arm l_h : _____ mm
Area S_h : _____ cm²
Elevator chord: _____ mm

$S_h = V_h S \bar{c} / l_h \cdot \text{start } V_h \approx 0.50\text{--}0.60$

VERTICAL TAIL

V_v : _____
Tail arm l_v : _____ mm
Area S_v : _____ cm²
Rudder chord: _____ mm

$S_v = V_v S b / l_v \cdot \text{start } V_v \approx 0.035\text{--}0.05$

CG + CONTROLS

CG datum: _____
Maiden CG: _____ mm / _____ % MAC
Ail: _____° Elev: _____° Rud: _____°

maiden at 10–15% static margin, low rates

POWER + RADIO

Battery: _____ S _____ mAh
Prop: _____
Measured A/W: _____ / _____
ESC/BEC: _____
Receiver/servos: _____

trainer $\approx$ 130–160 W/kg; verify with a wattmeter

AS BUILT — measured all-up mass: _____ g · measured CG: _____ mm aft of LE · date / builder: _____

Build-stage gates

Do not advance because the calendar says so. Advance when the evidence says the stage is complete.

DESIGN FREEZE

- ☐ **Mission written down**
One primary mission with numbers — target mass, stall speed, field size. Not adjectives.
- ☐ **Wing and tail sized**
Area, span, chord, and both tail volumes on the worksheet, sanity-checked against a proven model of similar size.
- ☐ **CG range and datum defined**
You know where the CG must sit, what you measure it from, and how the battery moves to trim it.
- ☐ **Power system chosen from data**
Motor, prop, ESC, and battery picked as a tested combination — not mixed from guesses.

STRUCTURE

- ☐ **Load paths identified**
You can point at the drawing and say how wing lift reaches the fuselage and where landing loads go.
- ☐ **Jigs and datums prepared**
Flat board, incidence blocks, and matched rib sets ready before any glue is opened.
- ☐ **Access designed in**
Battery, servos, receiver, and cooling openings planned now — not carved out of finished structure later.
- ☐ **Materials compatibility checked**
Adhesives tried on scrap of the actual wood or foam; hardware sized to the loads it carries.

BEFORE COVERING

- ☐ **Incidence and washout measured**
Both panels read the same numbers from the same datum — written down, not eyeballed.
- ☐ **Tail square, fin vertical**
Diagonal measurements match left and right; the fin is checked from the front and the rear.
- ☐ **Linkages cycle freely**
Full travel with no slop, no binding, and no servo buzz at the extremes.
- ☐ **Mass and CG updated**
Estimates replaced with weighed numbers — and the balance plan still works with them.

READY TO MAIDEN

- ☐ **Power verified**
Full-throttle current, watts, and temperatures measured with a wattmeter — cowling installed.
- ☐ **Battery, failsafe, range test**
Pack restrained for crash loads; failsafe closes the throttle; range test done per the radio manual.
- ☐ **Controls confirmed**
Correct directions checked from behind the plane, low rates selected, trims centered.
- ☐ **Maiden plan agreed**
Written flight plan and abort criteria, helper briefed, conditions chosen — before driving to the field.

Sources, validation, and the road ahead

Use authoritative sources, successful comparable models, measured test data, and disciplined flight notes together.

CORE REFERENCES

- NASA Glenn — Beginner's Guide to Aeronautics — lift, drag, forces, controls, interactive sims.
www1.grc.nasa.gov/beginners-guide-to-aeronautics
- MIT OpenCourseWare — Unified Engineering (16.01) — preliminary design and stability rules.
ocw.mit.edu/courses/16-01-unified-engineering...
- UIUC Airfoil Coordinates Database — published geometry, low-Re test data.
m-selig.ae.illinois.edu/ads/coord_database.html
- Academy of Model Aeronautics — safety code, education, technical library.
modelaircraft.org

VERIFY WITH

- Component manufacturer data sheets and tested motor/prop combinations.
- Local regulations, club rules, insurance requirements, and field procedures.
- A proven aircraft of similar size, configuration, Reynolds number, and mission.
- Measured mass, current, power, temperature, CG, geometry, and flight behavior.

PLANNED DEEP-DIVE VOLUMES

- Detailed aerodynamic analysis and XFOIL workflow.
- Advanced structures and composite techniques.
- Electric propulsion and propeller matching.
- Gliders, flying wings, biplanes, and unusual configurations.
- 3D-printed fixed wings: LW-PLA structures, print orientation, spar integration.
- CFD modelling: from XFLR5 panel methods to full Navier–Stokes — and when each is worth the effort.

BUILDER'S EVIDENCE PACK

- Design worksheet and mass budget.
- Dimensioned drawings and datums.
- Photos of critical joints and wiring.
- Ground-test results and maiden-flight log.

 The strongest design habit is intellectual honesty: label assumptions, measure finished values, record failures, and update the model instead of defending the original guess.

ONE SENTENCE TO REMEMBER

Size for the mission, balance conservatively, build straight and stiff, test at safe altitude, and change only one major variable at a time.